

Imbedding of Jordan Algebras Into Lie Algebras. II
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IMBEDDING OF JORDAN ALGEBRAS INTO LIE ALGEBRAS. II.

By MAX KOECHER.*

Introduction. In the sequel the algebras considered are algebras over a given field of characteristic different from two. In this second part we deal with pairs $(\mathfrak{A};J)$ where $\mathfrak{A}$ is a Jordan algebra (or more generally an admissible algebra) and where J is an automorphism of $\mathfrak{A}$ satisfying $J^2=\mathrm{Id}$. With such a pair we associate a Lie algebra $\mathfrak{G}=\mathfrak{G}(\mathfrak{A};J)$ that is isomorphic to a subalgebra of $\mathfrak{L}(\mathfrak{A})$ (see [6], referred as part I).

Any ideal of $\mathfrak{G}$ induces an "associator ideal" of $\mathfrak{A}$ and vice versa. A necessary and sufficient condition for $\mathfrak{G}$ to be simple is given in § 2. In § 3 we introduce the notion of "decomposing" automorphism which turns out to be basic for our paper. A new type of "associative" bilinear forms is introduced in § 4 and used in § 5 for an explicit description of the Killing form of $\mathfrak{G}$ whenever $\mathfrak{A}$ has finite dimension. For a finite dimensional semi simple Jordan algebra $\mathfrak{A}$ over a field of characteristic zero and a decomposing automorphism J we show that $\mathfrak{G}$ is a reductive Lie algebra. Important examples for decomposing automorphisms are given in § 6: the automorphisms J_c defined by idempotents c of $\mathfrak{A}$. We are able to describe in detail the associator ideals of a pair $(\mathfrak{A};J_c)$ for simple algebras of arbitrary dimension.

1. The algebras $\mathfrak{G}(\mathfrak{A};J)$.

1. We start with a pair $(\mathfrak{A};J)$, where $\mathfrak{A}$ is a commutative algebra with unit element e and where $J:\mathfrak{A}\to\mathfrak{A}$ is an automorphism of the algebra satisfying $J^2=\mathrm{Id}$. Suppose that $\mathfrak{A}$ is admissible in the sense of I, § 2. **1.** Hence there exists an antiautomorphism $T\to T^*$ of the Lie algebra

$$\mathfrak{h}(\mathfrak{A})=L(\mathfrak{A})+[L(\mathfrak{A}),L(\mathfrak{A})]+\cdots$$

which is the identity on $L(\mathfrak{A})$.

The automorphism J induces a direct sum decomposition $\mathfrak{A}=\mathfrak{A}_+ + \mathfrak{A}_-$, where $\mathfrak{A}_+$ resp. $\mathfrak{A}_-$ are the eigenspaces of J with respect to the eigenvalues $+1$ resp. -1. For $a\in\mathfrak{A}$ we write in an obvious manner $a=a_+ + a_-$. Note that $\mathfrak{A}_+$ is a subalgebra of $\mathfrak{A}$ and that $\mathfrak{A}_+$ acts on $\mathfrak{A}_-$.

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We consider the automorphism $T \to T^J = -JT^*J$ of the Lie algebra $\mathfrak{h}(\mathfrak{A})$ and the corresponding direct sum decomposition $\mathfrak{h}(\mathfrak{A}) = \mathfrak{h}^+ + \mathfrak{h}^-$ where

$$\mathfrak{h}^\pm = \mathfrak{h}^\pm(\mathfrak{A}; J) = \{T; T \in \mathfrak{h}(\mathfrak{A}), T^J = \pm T\}$$

(see I, § 2. **6**). Note that the automorphism $T \to -T^*$ maps both spaces $\mathfrak{h}^+$ and $\mathfrak{h}^-$ onto itself.

One verifies that $L(a)$ belongs to $\mathfrak{h}^\pm$ if and only if $a \in \mathfrak{A}_\mp$. Since $\mathfrak{h}(\mathfrak{A})$ is generated as an algebra by $L(a) = L(a_+) + L(a_-)$ the vector space $\mathfrak{h}^+$ is spanned by the Lie products of transformations $L(u)$, where $u \neq 0$ is in $\mathfrak{A}_+$ *or* in $\mathfrak{A}_-$ and where in each product an *even* number of elements out of $\mathfrak{A}_+$ occurs:

$$(1.1)\quad \mathfrak{h}^+ = \mathfrak{h}^+(\mathfrak{A}; J) = L(\mathfrak{A}_-) + [L(\mathfrak{A}_-), L(\mathfrak{A}_-)] + [L(\mathfrak{A}_+), L(\mathfrak{A}_+)] + \cdots.$$

By a similar argument we obtain

$$(1.2)\qquad \mathfrak{h}^- = \mathfrak{h}^-(\mathfrak{A}; J) = L(\mathfrak{A}_+) + [L(\mathfrak{A}_-), L(\mathfrak{A}_+)] + \cdots.$$

In § 7 we restrict ourselves to finite dimensional formal real Jordan algebras and we give some results about Cartan decompositions of the real Lie algebra $\mathfrak{G}$. In particular we prove a decomposition of $[\mathfrak{G}, \mathfrak{G}]$ that is characteristic for the Lie algebras of bounded symmetric domains.

For the results used for Jordan algebras see the forthcoming book of N. Jacobson [4] or (in the finite dimensional case) H. Braun and M. Koecher [1], from which we also quote results for arbitrary Jordan algebras provided the proof carries over to the general situation.

I am indebted to J. Tits for the information that the construction of the Lie algebra $\mathfrak{L}(\mathfrak{A})$ in part I is a special case of one of his constructions (see J. Tits [9]).

2. Using the pairing

$$a \square b = 2L(ab) + 2[L(a), L(b)]$$

(see I, § 2. **1**) we define two other pairings of $\mathfrak{A} \times \mathfrak{A}$ into $\mathfrak{h}(\mathfrak{A})$ by

$$a \triangle b = a \square Jb + b \square Ja = b \triangle a,$$

$$a \nabla b = a \square Jb - b \square Ja = -b \nabla a.$$

From $Ja \square Jb = J(a \square b)J$ and (I; 2.2) we obtain $(a \square Jb)^* = J(b \square Ja)J$ and consequently we have

$$(1.3)\qquad (a \triangle b)^* = J(a \triangle b)J = Ja \triangle Jb,$$

$$(1.4)\qquad (a \nabla b)^* = -J(a \nabla b)J = -Ja \nabla Jb.$$

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Hence $a \triangle b \in \mathfrak{h}^-$ *resp.* $a \nabla b \in \mathfrak{h}^+$ whenever $a, b \in \mathfrak{A}$. Furthermore, using the definition of the pairings one verifies

$$(1.5) \qquad \mathfrak{A} \triangle \mathfrak{A} = L(\mathfrak{A}_+) + [L(\mathfrak{A}_-), L(\mathfrak{A}_+)] \subset \mathfrak{h}^-,$$

$$(1.6) \qquad \mathfrak{A} \nabla \mathfrak{A} = L(\mathfrak{A}_-) + [L(\mathfrak{A}_-), L(\mathfrak{A}_-)] + [L(\mathfrak{A}_+), L(\mathfrak{A}_+)] \subset \mathfrak{h}^+$$

Remark. For a Jordan algebra $\mathfrak{A}$ combining (1.1) and (1.6) and using $\mathfrak{h}(\mathfrak{A}) = L(\mathfrak{A}) + [L(\mathfrak{A}), L(\mathfrak{A})]$ we obtain $\mathfrak{h}^+ = \mathfrak{A} \nabla \mathfrak{A}$.

3. Denote by $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J)$ the subspace of elements $g \in \mathfrak{A}$ such that

$$(1.7) \qquad \mathfrak{h}^+ g = 0 \text{ and } \mathfrak{A} \nabla g = 0$$

is satisfied.

LEMMA 1. *Let* $\mathfrak{A}$ *be an admissible algebra and let* J *be an automorphism of* $\mathfrak{A}$ *satisfying* $J^2 = \mathrm{Id}$. *Then for* $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J)$ *it follows*

(i) $\mathfrak{c} \subset \mathfrak{A}_+$, $(\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{c}) = (\mathfrak{A}_+, \mathfrak{c}, \mathfrak{A}_+) = 0$, *i.e.* $\mathfrak{c}$ *is contained in the center* $\mathfrak{Z}(\mathfrak{A}_+)$ *of* $\mathfrak{A}_+$,

(ii) $\mathfrak{c}\mathfrak{A}_- = 0$,

(iii) $[L(\mathfrak{A}_+), L(\mathfrak{c})] = 0$.

Proof. Applying the second condition of (1.7) on e we obtain $a \cdot Jg = g \cdot Ja$ for $a \in \mathfrak{A}$ and consequently $g \in \mathfrak{A}_+$ as well as $g\mathfrak{A}_- = 0$. For $x, y \in \mathfrak{A}_+$ we conclude $[L(x), L(y)]g = (x \nabla g)y = 0$ from (1.7) and (1.1). Hence (i) is a consequence of (1.7). But $(\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{c}) = 0$ together with $(\mathfrak{A}_+, \mathfrak{A}_-, \mathfrak{c}) = 0$ implies (iii).

Remark. For a Jordan algebra we see that $g \in \mathfrak{c}$ is equivalent to $g \in \mathfrak{A}_+$, $(\mathfrak{A}_+, \mathfrak{A}_+, g) = (\mathfrak{A}_+, g, \mathfrak{A}_+) = 0$, i.e. $g \in \mathfrak{Z}(\mathfrak{A}_+)$ and $g\mathfrak{A}_- = 0$.

4. Next we define in the vector space

$$\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J) = \mathfrak{h}^+(\mathfrak{A}; J) \oplus \mathfrak{A}$$

an anticommutative product $[x_1, x_2] = T \oplus a$, where

$$T = [T_1, T_2] + a_1 \nabla a_2, \quad a = T_1 a_2 - T_2 a_1,$$

whenever $x_i = T_i \oplus a_i \in \mathfrak{G}$. Obviously, the space $\mathfrak{G}$ is closed under this product. *Hence* $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ *is an anticommutative algebra having* $\mathfrak{h}^+ = \mathfrak{h}^+(\mathfrak{A}; J)$ *as subalgebra.* Moreover one obtains $[\mathfrak{h}^+, \mathfrak{A}] \subset \mathfrak{A}$ and $[\mathfrak{A}, \mathfrak{A}] = \mathfrak{A} \nabla \mathfrak{A} \subset \mathfrak{h}^+$. We identify the spaces $\mathfrak{h}^+$ and $\mathfrak{A}$ with the corresponding subspaces of $\mathfrak{G}$.

There are three standard automorphism Θ_ϵ, $\epsilon = 0, \pm 1$, of the algebra $\mathfrak{G}$ defined by

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$$\Theta_0 x = T \oplus (-a) \text{ and } \Theta_\epsilon x = (-T^*) \oplus \epsilon Ja \text{ for } \epsilon = \pm 1,$$

whenever $x = T \oplus a \in \mathfrak{G}$. We write Θ_+ resp. Θ_- for Θ_ϵ, $\epsilon = \pm 1$. Using (1.4) and the fact that $T \to -T^*$ is an automorphism of $\mathfrak{h}^+$ one verifies *that* Θ_ϵ *are automorphisms of the algebra* $\mathfrak{G}$ *for* $\epsilon = 0, \pm 1$, *satisfying*

$$\Theta_+\Theta_- = \Theta_-\Theta_+ = \Theta_0, \qquad \Theta_\epsilon^2 = \mathrm{Id} \text{ for } \epsilon = 0, \pm 1,$$

$$\Theta_\epsilon\Theta_0 = \Theta_0\Theta_\epsilon = \Theta_{-\epsilon} \text{ for } \epsilon = \pm 1.$$

Denote by $\mathfrak{C} = \mathfrak{C}(\mathfrak{A}; J)$ the center of the algebra $\mathfrak{G}$, i.e. the set of elements $z \in \mathfrak{G}$ satisfying $[z, x] = 0$ for all $x \in \mathfrak{G}$. One verifies that $\mathfrak{C}$ coincides with $\mathfrak{c} = 0 \oplus \mathfrak{c}$, where $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J)$ is the subspace of $\mathfrak{A}$ defined in **3**.

From the definition of the product in $\mathfrak{G}$ it follows

$$[\mathfrak{G}, \mathfrak{G}] = ([\mathfrak{h}^+, \mathfrak{h}^+] + \mathfrak{A} \nabla \mathfrak{A}) \oplus \mathfrak{h}^+\mathfrak{A}.$$

Remark. For a Jordan algebra $\mathfrak{A}$ we obtain $[\mathfrak{G}, \mathfrak{G}] = \mathfrak{h}^+ \oplus \mathfrak{h}^+\mathfrak{A}$.

5. Besides $\mathfrak{G}$ we consider also the algebra $\mathfrak{L}(\mathfrak{A}) = \mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{A} \oplus \mathfrak{A}$. One has to distinguish carefully between the elements $T \oplus a$ of $\mathfrak{G}$ and $T \oplus a \oplus 0$ of $\mathfrak{L}(\mathfrak{A})$ because the product of two such elements is not the same in these algebras. However, $\mathfrak{G}$ can be considered as a subalgebra of $\mathfrak{L}(\mathfrak{A})$ as follows:

LEMMA 2. *Let* $\mathfrak{A}$ *be an admissible algebra and let* J *be an automorphism of* $\mathfrak{A}$ *satisfying* $J^2 = \mathrm{Id}$. *Then the injection* $\pi: \mathfrak{G} \to \mathfrak{L}(\mathfrak{A})$ *given by* $\pi(T \oplus a) = T \oplus a \oplus \overline{Ja}$ *is a homomorphism of the algebras.*

Proof. First of all π is an injection of $\mathfrak{G}$ into $\mathfrak{L}(\mathfrak{A})$. Calculating the product $[\pi(T_1 \oplus a_1), \pi(T_2 \oplus a_2)] = T \oplus a \oplus \bar{b}$ in the algebra $\mathfrak{L}(\mathfrak{A})$ we find

$$T = [T_1, T_2] + a_1 \square Ja_2 - a_2 \square Ja_1 = [T_1, T_2] + a_1 \nabla a_2$$

and

$$b = T_2^* Ja_1 - T_1^* Ja_2 = J(T_1 a_2 - T_2 a_1) = Ja.$$

Hence this product coincides with the images of $[T_1 \oplus a_1, T_2 \oplus a_2]$ under π.

Using the subspaces

$$\mathfrak{L}^\pm(\mathfrak{A}; J) = \{T \oplus a \oplus (\pm Ja); T \in \mathfrak{h}^\pm, a \in \mathfrak{A}\}$$

of $\mathfrak{L}(\mathfrak{A})$ (see I, § 2. **6**) we obtain

COROLLARY 1. $\mathfrak{G}$ *is isomorphic to* $\mathfrak{L}^+(\mathfrak{A}; J)$.

Since $\mathfrak{L}^+(\mathfrak{A}; J)$ is a subalgebra of $\mathfrak{L}(\mathfrak{A})$, we obtain from I, Theorem 2,

COROLLARY 2. $\mathfrak{G}$ *is a Lie algebra whenever* $\mathfrak{A}$ *is Jordan.*

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On the other hand we have

LEMMA 3. *Let $\mathfrak{A}$ be an admissible algebra and let J be an automorphism of $\mathfrak{A}$ satisfying $J^2 = \mathrm{Id}$. Then $\mathfrak{G}$ is a Lie algebra if and only if $[T, a \nabla b] = Ta \nabla b + a \nabla Tb$ for $a, b \in \mathfrak{A}$ and $T \in \mathfrak{h}^+$.*

Proof. A similar argument as used for the proof of I, Lemma 4, shows that $\mathfrak{G}$ is a Lie algebra if and only if the condition stated in the Lemma and $(a \nabla b)c + (b \nabla c)a + (c \nabla a)b = 0$ are satisfied. But the last condition is a consequence of (I; 2.2).

It would be of interest to know if there is an admissible algebra $\mathfrak{A}$, which is not a Jordan algebra but for which $\mathfrak{G}$ is a Lie algebra for some automorphism J.

6. Since $\mathfrak{L}^+(\mathfrak{A}; J)$ acts faithfully on $\mathfrak{L}^-(\mathfrak{A}; J)$ (see I, § 2.**6**) we are able to define a module for $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ in the following way: We define an action of $\mathfrak{G}$ in the vector space

$$\mathfrak{M} = \mathfrak{M}(\mathfrak{A}; J) = \mathfrak{h}^-(\mathfrak{A}; J) \oplus \mathfrak{A}$$

by setting

$$(1.8) \qquad [x, y] = [T \oplus a, S \oplus b] = ([T, S] - a \triangle b) \oplus (Tb - Sa)$$

whenever $x = T \oplus a \in \mathfrak{G}$ and $y = S \oplus b \in \mathfrak{M}$. Here the symmetric pairing $a \triangle b$ is defined in **2**. From (1.3) it follows that the right hand side of (1.8) belongs to $\mathfrak{M}$. Now one can verify again *that $\mathfrak{G}$ acts faithfully on $\mathfrak{M}$*. In particular it follows that $\mathfrak{G}$ is a Lie algebra whenever $\mathfrak{M}$ is a Lie module for $\mathfrak{G}$. Again one verifies

LEMMA 4. *Let $\mathfrak{A}$ be an admissible algebra and let J be an automorphism of $\mathfrak{A}$ satisfying $J^2 = \mathrm{Id}$. Then $\mathfrak{M}$ is a Lie module for $\mathfrak{G}$ if and only if*

$$[T, a \triangle b] = Ta \triangle b + a \triangle Tb, \; T \in \mathfrak{h}^+,$$
$$[S, a \nabla b] = Sa \triangle b - a \triangle Sb, \; S \in \mathfrak{h}^-,$$

and $a, b \in \mathfrak{A}$.

Since for a Jordan algebra $\mathfrak{A}$ we have $[T, a \square b] = Ta \square b - a \square T^* b$, $T \in \mathfrak{h}(\mathfrak{A})$ (see (I; 1.10) and I, § 3.**1**) both conditions are satisfied. Therefore we have

COROLLARY. *If $\mathfrak{A}$ is a Jordan algebra, then $\mathfrak{M}$ is a Lie module for $\mathfrak{G}$.*

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Without proof we mention two further results about the action on $\mathfrak{G}$ on $\mathfrak{M}$: For $x \in \mathfrak{G}$, $y \in \mathfrak{M}$, put $R(x)y = [x, y]$ and $R(\mathfrak{G}) = \{R(x); x \in \mathfrak{G}\}$. Then $x \to R(x)$ is a faithful representation of $\mathfrak{G}$ in $\mathrm{Hom}(\mathfrak{M}, \mathfrak{M})$. There is an imbedding of the center $\mathfrak{Z}(\mathfrak{A})$ of $\mathfrak{A}$ and of $\mathfrak{c}$ into $\mathrm{Hom}(\mathfrak{M}, \mathfrak{M})$, namely

$$A_f y = L(f)S \oplus fb, \quad f \in \mathfrak{Z}(\mathfrak{A}),$$
$$B_g y = b \triangle g \oplus Sg, \quad g \in \mathfrak{c}.$$

Let $\mathfrak{A}$ be a finite dimensional semi simple Jordan algebra. Then the centralizer of $R(\mathfrak{G})$ in $\mathrm{Hom}(\mathfrak{M}, \mathfrak{M})$ is spanned by A_f, $f \in \mathfrak{Z}(\mathfrak{A})$, and B_g, $g \in \mathfrak{c}$. Moreover there exists an antiautomorphism $Z \to Z^{\#}$ of $\mathrm{Hom}(\mathfrak{M}, \mathfrak{M})$ such that $R^{\#}(x) = -R(x)$ for $x \in \mathfrak{G}$. Hence $R(\mathfrak{G})$ is a subalgebra of the Lie algebra of elements Z fulfilling $Z^{\#}A_f = -A_f Z$, $Z^{\#}B_g = -B_g Z$, for $f \in \mathfrak{Z}(\mathfrak{A})$ and $g \in \mathfrak{c}$.

7. According to Corollary 1 of Lemma 2 the algebras $\mathfrak{G}$ are isomorphic to proper subalgebras of $\mathfrak{L}(\mathfrak{A})$. But taking another algebra $\mathfrak{A}'$ one gets $\mathfrak{L}(\mathfrak{A})$ as $\mathfrak{G}(\mathfrak{A}'; J)$: For a better distinction we denote the elements of the direct sum $\mathfrak{A} \oplus \mathfrak{A}$ by (u, v). Let J be the canonical automorphism of the algebra $\mathfrak{A} \oplus \mathfrak{A}$ defined by $J(u, v) = (v, u)$. Since the left multiplication of $\mathfrak{A} \oplus \mathfrak{A}$ is the sum of the left multiplications of $\mathfrak{A} \oplus 0$ and of $0 \oplus \mathfrak{A}$ we may identify $\mathfrak{h}(\mathfrak{A} \oplus \mathfrak{A})$ and the algebra $\mathfrak{h}(\mathfrak{A}) \oplus \mathfrak{h}(\mathfrak{A})$. Using this identification we have for a pair (T, S) in $\mathfrak{h}(\mathfrak{A} \oplus \mathfrak{A})$ obviously $(T, S)^* = (T^*, S^*)$ and $(T, S)J = J(S, T)$. Hence the automorphism of $\mathfrak{h}(\mathfrak{A} \oplus \mathfrak{A})$ induced by J is given by $(T, S)^J = -J(T, S)^* J = (-S^*, -T^*)$. We see that (T, S) belongs to $\mathfrak{h}^+(\mathfrak{A} \oplus \mathfrak{A}; J)$ if and only if $S = -T^*$. Therefore the map

$$\psi: \mathfrak{h}(\mathfrak{A}) \to \mathfrak{h}^+(\mathfrak{A} \oplus \mathfrak{A}; J), \quad \psi(T) = (T, -T^*),$$

is a bijection. One verifies *that ψ is an isomorphism of the Lie algebras.*

Next we extend the map ψ to a map of $\mathfrak{L}(\mathfrak{A})$ into $\mathfrak{G}(\mathfrak{A} \oplus \mathfrak{A}; J)$ by setting $\psi(T \oplus a \oplus \bar{b}) = \psi(T) \oplus (a, b)$. Using $(a_1, b_1) \nabla (a_2, b_2) = \psi(a_1 \square b_2 - a_2 \square b_1)$ one verifies that $\psi: \mathfrak{L}(\mathfrak{A}) \to \mathfrak{G}(\mathfrak{A} \oplus \mathfrak{A}; J)$ is a homomorphism of the algebras. Summing up we have

LEMMA 5. *Let $\mathfrak{A}$ be an admissible algebra and let J be the canonical automorphism of $\mathfrak{A} \oplus \mathfrak{A}$. Then $\psi: \mathfrak{L}(\mathfrak{A}) \to \mathfrak{G}(\mathfrak{A} \oplus \mathfrak{A}; J)$ is an isomorphism of the algebras.*

2. The ideals of $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$.

1. Let $\mathfrak{A}$ be an admissible algebra and J, $J^2 = \mathrm{Id}$, be an automorphism

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of $\mathfrak{A}$. For a subspace $\mathfrak{m}$ of $\mathfrak{A}$ we denote by $\mathfrak{i}^+(\mathfrak{m})$ the ideal in $\mathfrak{h}^+ = \mathfrak{h}^+(\mathfrak{A}; J)$ generated by $\mathfrak{A} \nabla \mathfrak{m}$. Setting

$$\mathfrak{m}_+ = \{Jm + m; m \in \mathfrak{m}\} \text{ and } \mathfrak{m}_- = \{Jm - m; m \in \mathfrak{m}\}$$

we obtain $\mathfrak{m} \subset \mathfrak{m}_+ + \mathfrak{m}_-$ but in general $\mathfrak{m}_+$ is not contained in $\mathfrak{m}$. This is the case if and only if $J\mathfrak{m} \subset \mathfrak{m}$. From the definition of $a \nabla b$ it follows

$$\tfrac{1}{2} a \nabla b = L(a \cdot Jb - b \cdot Ja) + [L(a), L(Jb)] + [L(Ja), L(b)]$$

and consequently

$$(2.1) \quad \tfrac{1}{4} a \nabla b = L(a_- b_+ - a_+ b_-) + [L(a_+), L(b_+)] - [L(a_-), L(b_-)].$$

Using the subspace $\mathfrak{i}(\mathfrak{m})$ of $\mathfrak{h}(\mathfrak{A})$ defined in I, § 2. **3**, we obtain $\mathfrak{i}^+(\mathfrak{m}) \subset \mathfrak{h}^+ \cap \mathfrak{i}(\mathfrak{m})$.

On the other hand we define the subspace

$$\mathfrak{j}^+(\mathfrak{m}) = \{T; T \in \mathfrak{h}^+, T\mathfrak{A} \subset \mathfrak{m}\}$$

of $\mathfrak{h}^+$. One verifies *that* $\mathfrak{j}^+(\mathfrak{m})$ *is an ideal in* $\mathfrak{h}^+$ *whenever* $\mathfrak{h}^+\mathfrak{m} \subset \mathfrak{m}$. Furthermore, $J\mathfrak{m} = \mathfrak{m}$ implies that $\mathfrak{j}^+(\mathfrak{m})$ is mapped into itself under the map $T \to T^*$. Using the subspace $\mathfrak{j}(\mathfrak{m})$ of $\mathfrak{h}(\mathfrak{A})$ defined in I, § 2. 3, we obtain $\mathfrak{j}(\mathfrak{m}) = \mathfrak{h}^+ \cap \mathfrak{j}(\mathfrak{m})$ whenever $J\mathfrak{m} = \mathfrak{m}$.

2. A subspace $\mathfrak{m}$ of $\mathfrak{A}$ is called an *associator ideal of the pair* $(\mathfrak{A}; J)$ if

$$(2.2) \qquad \mathfrak{h}^+\mathfrak{m} \subset \mathfrak{m} \text{ and } (\mathfrak{A} \nabla \mathfrak{m})\mathfrak{A} \subset \mathfrak{m}$$

is satisfied. Note that any ideal of $\mathfrak{A}$ satisfying $J\mathfrak{m} = \mathfrak{m}$ is also an associator ideal of $(\mathfrak{A}; J)$. We conclude from (1.7) *that the subspace* $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J)$ *of* $\mathfrak{A}$ *is an associator ideal of* $(\mathfrak{A}; J)$.

Let $\mathfrak{m}$ be an associator ideal of $(\mathfrak{A}; J)$. From $e \nabla m = L(Jm - m)$ it follows $Jm - m \in \mathfrak{m}$ for $m \in \mathfrak{m}$ and consequently $J\mathfrak{m} \subset \mathfrak{m}$. Hence $\mathfrak{m} = \mathfrak{m}_+ + \mathfrak{m}_-$ where $\mathfrak{m}_+ = \mathfrak{m} \cap \mathfrak{A}_+$ and $\mathfrak{m}_- = \mathfrak{m} \cap \mathfrak{A}_-$. Using (1.1) and (2.1) one verifies *that*

$$(2.3) \quad \mathfrak{A}_-\mathfrak{m} \subset \mathfrak{m}, \quad \mathfrak{A}\mathfrak{m}_- \subset \mathfrak{m}, \quad (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{m}_+) \subset \mathfrak{m}, \quad (\mathfrak{A}_+, \mathfrak{m}_+, \mathfrak{A}_+) \subset \mathfrak{m},$$

is satisfied whenever $\mathfrak{m}$ *is an associator ideal of* $(\mathfrak{A}; J)$.

In order to prove $\mathfrak{i}^+(\mathfrak{m}) \subset \mathfrak{j}^+(\mathfrak{m})$ it is enough to show $\mathfrak{A} \nabla \mathfrak{m} \subset \mathfrak{j}^+(\mathfrak{m})$. But this is the second condition of (2.2).

Remark. For a Jordan algebra $\mathfrak{A}$ a subspace $\mathfrak{m}$ is an associator ideal of

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$(\mathfrak{A};J)$ if and only if $J\mathfrak{m}\subset\mathfrak{m}$ and (2.3) are satisfied. In case that J is the identity the space $\mathfrak{m}$ is an associator ideal of $\mathfrak{A}$ in the sense of N. Jacobson [4].

3. Next we consider ideals of $\mathfrak{G}=\mathfrak{G}(\mathfrak{A};J)$ of the special form $\mathfrak{J}=\mathfrak{k}\oplus\mathfrak{m}$, where $\mathfrak{k}$ resp. $\mathfrak{m}$ are subspaces of $\mathfrak{h}^+$ resp. $\mathfrak{A}$. From the definition of the product in the algebra $\mathfrak{G}$ it follows *that* $\mathfrak{J}=\mathfrak{k}\oplus\mathfrak{m}$ *is an ideal in* $\mathfrak{G}$ *if and only if*

$$[\mathfrak{h}^+,\mathfrak{k}]\subset\mathfrak{k},\quad \mathfrak{h}^+\mathfrak{m}\subset\mathfrak{m},\quad \mathfrak{k}\mathfrak{A}\subset\mathfrak{m},\quad \mathfrak{A}\nabla\mathfrak{m}\subset\mathfrak{k}, \tag{2.4}$$

are satisfied. Combining (2.2) and 2.4) we obtain

LEMMA 6. $\mathfrak{J}=\mathfrak{k}\oplus\mathfrak{m}$ *is an ideal of* $\mathfrak{G}$ *if and only if*

(i) $\mathfrak{m}$ *is an associator ideal of the pair* $(\mathfrak{A};J)$ *and*

(ii) $\mathfrak{k}$ *is an ideal in* $\mathfrak{h}^+$ *such that* $\mathfrak{i}^+(\mathfrak{m})\subset\mathfrak{k}\subset\mathfrak{j}^+(\mathfrak{m})$.

As a consequence we see that the subspaces

$$\tilde{\mathfrak{J}}^+(\mathfrak{m})=\mathfrak{i}^+(\mathfrak{m})\oplus\mathfrak{m}\quad\text{and}\quad\mathfrak{J}^+(\mathfrak{m})=\mathfrak{j}^+(\mathfrak{m})\oplus\mathfrak{m}$$

are ideals in $\mathfrak{G}$ if and only if $\mathfrak{m}$ is an associator ideal of $(\mathfrak{A};J)$.

We define the projections

$$H:\mathfrak{G}\to\mathfrak{h}^+,\qquad A:\mathfrak{G}\to\mathfrak{A},$$

of the vector space by $Hx=T$ resp. $Ax=a$ whenever $x=T\oplus a$. Let $\mathfrak{J}$ and $\mathfrak{m}=A\mathfrak{J}$ one verifies that $\mathfrak{k}=H\mathfrak{J}$be an arbitrary ideal of $\mathfrak{G}$. Setting (2.4) is satisfied. Hence

$$\hat{\mathfrak{J}}=H\mathfrak{J}\oplus A\mathfrak{J}=\mathfrak{k}\oplus\mathfrak{m}$$

is an ideal of $\mathfrak{G}$, too, and we obtain the

COROLLARY. *Let* $\mathfrak{J}$ *be an ideal of* $\mathfrak{G}$. *Then* $\mathfrak{J}\subset\mathfrak{J}^+(\mathfrak{m})$ *where* $\mathfrak{m}=A\mathfrak{J}$ *is an associator ideal of* $(\mathfrak{A};J)$.

LEMMA 7. *Let* $\mathfrak{J}$ *be an ideal of* $\mathfrak{G}$ *such that* $\mathfrak{J}\cap\mathfrak{A}=0$ *and* $A\mathfrak{J}=\mathfrak{A}$. *Then there exists a linear injection* $S:\mathfrak{A}\to\mathfrak{h}^+$ *such that* $\mathfrak{J}=\{S(a)\oplus a;a\in\mathfrak{A}\}$ *and* $\mathfrak{A}\nabla\mathfrak{A}\subset S(\mathfrak{A})$.

Proof. From $(\mathfrak{J}\cap\mathfrak{h}^+)\mathfrak{A}=[\mathfrak{J}\cap\mathfrak{h}^+,\mathfrak{A}]\subset\mathfrak{J}\cap\mathfrak{A}=0$ we obtain $\mathfrak{J}\cap\mathfrak{h}^+=0$. Hence for $T\oplus a\in\mathfrak{J}$ the element T of $\mathfrak{h}^+$ is uniquely determined by a. Using $A\mathfrak{J}=\mathfrak{A}$ we see that an injection $S:\mathfrak{A}\to\mathfrak{h}^+$ exists. Finally for $a,b\in\mathfrak{A}$ it follows $a\nabla b\oplus S(a)b=[S(a)\oplus a,b]\in\mathfrak{J}$ and consequently $a\nabla b\in S(\mathfrak{A})$.

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Let $\mathfrak{B}$ be an arbitrary algebra and let Φ be an automorphism of $\mathfrak{B}$ satisfying $\Phi^2 = \mathrm{Id}$. Then the pair $(\mathfrak{B}; \Phi)$ is called *simple*, if $\mathfrak{B}$ has no proper ideal that is invariant under Φ. For a simple pair $(\mathfrak{B}; \Phi)$ only the following two types are possible:

(I) $\mathfrak{B}$ is isomorphic to $\mathfrak{C} \oplus \mathfrak{C}$, where $\mathfrak{C}$ is a simple ideal and Φ corresponds to the canonical automorphism of $\mathfrak{C} \oplus \mathfrak{C}$ that is given by $a \oplus b \to b \oplus a$,

(II) $\mathfrak{B}$ is a simple algebra.

Indeed, if $\mathfrak{b}$ is a proper ideal of $\mathfrak{B}$ then $\mathfrak{b} \cap \Phi\mathfrak{b}$ and $\mathfrak{b} + \Phi\mathfrak{b}$ are Φ-invariant ideals. Hence $\mathfrak{b} \cap \Phi\mathfrak{b} = 0$ and $\mathfrak{b} + \Phi\mathfrak{b} = \mathfrak{B}$. Hence $\mathfrak{b}$ is simple because an ideal of $\mathfrak{b}$ turns out to be an ideal of $\mathfrak{B}$.

4. We restrict ourselves now to the case that $\mathfrak{A}$ is a Jordan algebra. Linearization of the basic identity $(u, v, u^2) = 0$ leads to

$$(A') \qquad (a, b, cd) = (ac, b, d) + (ad, b, c), \quad a, b, c, d \in \mathfrak{A}.$$

Application of the identity (B) of I, § 3. **1** on d yields

$$(B') \qquad (a, cd, b) = c(a, d, b) + d(a, c, b), \quad a, b, c, d \in \mathfrak{A}.$$

The choice $a, b \in \mathfrak{A}_+$ and $c, d \in \mathfrak{A}_-$ leads to

$$(2.5) \qquad (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_-\mathfrak{A}_-) \subset \mathfrak{A}_-\mathfrak{A}_-, \quad (\mathfrak{A}_+, \mathfrak{A}_-\mathfrak{A}_-, \mathfrak{A}_+) \subset \mathfrak{A}_-\mathfrak{A}_-.$$

Setting

$$\mathfrak{A}'' = \mathfrak{A}\mathfrak{A}_- = \mathfrak{A}_- + \mathfrak{A}_-\mathfrak{A}_-$$

we show *that* $\mathfrak{A}''$ *is an associator ideal of* $(\mathfrak{A}; J)$. Using the above remark we have to prove only (2.3) for $\mathfrak{m} = \mathfrak{A}''$. But the multiplication rules between the spaces $\mathfrak{A}_+$ and $\mathfrak{A}_-$ reduce the conditions to (2.5).

The pair $(\mathfrak{A}; J)$ is called *associator simple* if 0 and $\mathfrak{A}$ are the only associator ideals of $(\mathfrak{A}; J)$. Since $\mathfrak{c}$ and $\mathfrak{A}''$ are associator ideals of $(\mathfrak{A}; J)$ we obtain $\mathfrak{c} = 0$ and $\mathfrak{A}'' = \mathfrak{A}$ or $\mathfrak{A}'' = 0$ for an associator simple pair.

Using the automorphism Θ_0 of the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ (see § 1. **4**) we prove

THEOREM 1. *Let* $\mathfrak{A}$ *be a Jordan algebra with unit element and let* $J \neq \mathrm{Id}$ *be an automorphism of* $\mathfrak{A}$ *satisfying* $J^2 = \mathrm{Id}$. *Then the pair* $(\mathfrak{G}; \Theta_0)$ *is simple if and only if the pair* $(\mathfrak{A}; J)$ *is associator simple.*

Moreover, if the pair $(\mathfrak{A}; J)$ *is associator simple and if* $\mathfrak{A}_+$ *acts irreducibly on* $\mathfrak{A}_-$ *then* $\mathfrak{G}$ *is simple.*

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Proof. Since $\mathfrak{J}^+(\mathfrak{m})$ is a proper ideal in $\mathfrak{G}$ whenever $\mathfrak{m}$ is a proper associator ideal of $(\mathfrak{A};J)$ and since $\mathfrak{J}^+(\mathfrak{m})$ is invariant under Θ_0, we see that $(\mathfrak{A};J)$ is associator simple if $(\mathfrak{G};\Theta_0)$ is simple.

Conversely suppose that $(\mathfrak{A};J)$ is associator simple. Since $J\neq \mathrm{Id}$ implies $\mathfrak{A}_-\neq 0$ we obtain $\mathfrak{A}''=\mathfrak{A}$. Now let $\mathfrak{J}$ be an ideal of $\mathfrak{G}$ that is invariant under Θ_0. Then $\mathfrak{J}$ splits, $\mathfrak{J}=\mathfrak{k}+\mathfrak{m}$, and $\mathfrak{m}$ is an associator ideal of $(\mathfrak{A};J)$. Now $\mathfrak{m}=0$ resp. $\mathfrak{m}=\mathfrak{A}$ yields $\mathfrak{J}=0$ resp. $\mathfrak{J}=\mathfrak{G}$.

For the remaining case let $\mathfrak{J}\neq 0$ be an arbitrary ideal of $\mathfrak{G}$ and let $\mathfrak{m}=A\mathfrak{J}$. The Corollary of Lemma 6 leads to $\mathfrak{J}\subset\mathfrak{J}^+(\mathfrak{m})$ and $\mathfrak{m}$ is an associator ideal of $(\mathfrak{A};J)$. We conclude $\mathfrak{m}=\mathfrak{A}$ because $\mathfrak{m}=0$ would imply $\mathfrak{J}^+(\mathfrak{m})=0$. Hence $A\mathfrak{J}=\mathfrak{A}$.

Furthermore we may assume $\mathfrak{J}\cap\mathfrak{A}=0$. Indeed, $\mathfrak{J}\cap\mathfrak{A}$ turns out to be an associator ideal of $(\mathfrak{A};J)$ and $\mathfrak{J}\cap\mathfrak{A}=\mathfrak{A}$ leads to $\mathfrak{J}=\mathfrak{G}$.

Now Lemma 7 shows that there is a bijection $S:\mathfrak{A}\to\mathfrak{h}^+=\mathfrak{A}\nabla\mathfrak{A}$ such that $\mathfrak{J}=\{S(a)\oplus a;a\in\mathfrak{A}\}$. Since $\mathfrak{J}$ is supposed to be an ideal we obtain

$$(*)\qquad a\nabla b=S(S(a)b),\quad S(Ta)=[T,S(a)]\ \text{for}\ a,b\in\mathfrak{A},T\in\mathfrak{h}^+,$$

and consequently

$$[S(a),S(b)]=a\nabla b\ \text{for}\ a,b\in\mathfrak{A}.$$

Setting $Q=-S(e)$ from (*) we obtain $S(Qe)=-S(S(e)e)=-e\nabla e=0$ and consequently $Q\in\mathfrak{h}^+\cap[L(\mathfrak{A}),L(\mathfrak{A})]$. Hence (*) yields

$$S(a)=L(Qa),\quad S(ab)=[L(a),L(Qb)]\ \text{for}\ a,b\in\mathfrak{A}_-.$$

Next $4L(a)=a\nabla e=[Q,S(a)]=L(Q^2a)$ leads to

$$Q^2a=4a\ \text{for}\ a\in\mathfrak{A}_-.$$

Let $a,b,c\in\mathfrak{A}_-$. Again using (*) we obtain

$$L(Q[a\cdot bc])=S(a\cdot bc)=[L(a),S(bc)]=[L(a),[L(b),L(Qc)]]$$

and therefore

$$Q(a\cdot bc)=Qc\cdot ab-b(a\cdot Qc).$$

Interchanging b and c and replacing b,c by Qb,Qc we obtain

$$(**)\qquad a(Qb\cdot Qc)=-4a\cdot bc.$$

Hence $a(b\cdot Qc)=-a(Qb\cdot c)$ and

$$\begin{aligned}-4Q(a\cdot bc)&=Q[a(Qb\cdot Qc)]\\&=Qa(Qb\cdot Qc)+4a(b\cdot Qc)+4a(Qb\cdot c)=Qa(Qb\cdot Qc)=-4Qa\cdot bc.\end{aligned}$$

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From $\mathfrak{A} = \mathfrak{A}''$ we conclude $\mathfrak{A}_-\mathfrak{A}_- = \mathfrak{A}_+$ and therefore

$$Q(wa) = w(Qa) \text{ for } w \in \mathfrak{A}_+ \text{ and } a \in \mathfrak{A}_-.$$

Consequently the eigenspaces $\mathfrak{B}_\epsilon = \{a; a \in \mathfrak{A}_-, Qa = 2\epsilon a\}$, $\epsilon = \pm 1$, are invariant under the action of $\mathfrak{A}_+$. The last condition of the theorem yields $\mathfrak{B}_1 = \mathfrak{A}_-$ or $\mathfrak{B}_{-1} = \mathfrak{A}_-$. In both cases we conclude $\mathfrak{A}_+\mathfrak{A}_- = 0$ from (**) which is impossible.

5. Since any J-invariant ideal of $\mathfrak{A}$ is also an associator ideal of the pair $(\mathfrak{A}; J)$ we conclude that $(\mathfrak{A}; J)$ *is a simple pair whenever* $(\mathfrak{A}; J)$ *is associator simple.* Hence we have two types (I) and (II) of **3.**

We prove *that any simple pair* $(\mathfrak{A}; J)$ *of type* I *is associator simple.* An application of Lemma 5 shows that $\mathfrak{G}$ and $\mathfrak{L}(\mathfrak{B})$, $\mathfrak{A} = \mathfrak{B} \oplus \mathfrak{B}$, are isomorphic. Therefor $\mathfrak{G}$ is simple because of I, Theorem 1, and consequently $(\mathfrak{A}; J)$ is associator simple because of Theorem 1.

A simple pair of type II is not necessarily associator simple, but there is a sufficient condition as follows:

Suppose that $\mathfrak{A}_+$ *acts faithfully and irreducibly on* $\mathfrak{A}_-$ *and suppose that* $\mathfrak{A}_-\mathfrak{A}_- = \mathfrak{A}_+$ *is fulfilled. Then* $(\mathfrak{A}; J)$ *is associator simple, and moreover* $\mathfrak{G}$ *is simple* (see Theorem 1).

Indeed, if $\mathfrak{m} = \mathfrak{m}_+ + \mathfrak{m}_-$ is a given associator ideal of $(\mathfrak{A}; J)$ then $\mathfrak{A}_+$ acts on the subspace $\mathfrak{m}_-$ of $\mathfrak{A}_-$. Hence $\mathfrak{m}_- = \mathfrak{A}_-$ or $\mathfrak{m}_- = 0$. The first case leads to $\mathfrak{A}_+ = \mathfrak{A}_-\mathfrak{A}_- \subset \mathfrak{m}_+$, i.e. to $\mathfrak{m} = \mathfrak{A}$, and the second case to $\mathfrak{m}_+\mathfrak{A}_- = 0$, i.e. to $\mathfrak{m} = 0$ because $\mathfrak{A}_+$ acts faithfully.

For example let $K(\vartheta)$ be a quadratic extension of the field K and let $\alpha \to \bar{\alpha}$ be the canonical automorphism of $K(\vartheta)$ over K. We consider the Jordan algebra $\mathfrak{A}$ of hermitian $r \times r$ matrices $a = (\alpha_{ij})$, $(\alpha_{ij}) = \alpha_{ji} \in K(\vartheta)$, with automorphism J defined by $Ja = (\alpha_{ji})$. Then $\mathfrak{A}_+$ resp. $\mathfrak{A}_-$ consists of the symmetric resp. skew symmetric matrices with entries in K resp. $\vartheta \cdot K$. It is known (see N. Jacobson [5]) that $\mathfrak{A}_-$ is irreducible under the action of $\mathfrak{A}_+$. One verifies for $r > 2$ by induction on r that $\mathfrak{A}_+$ acts faithfully and that $\mathfrak{A}_-\mathfrak{A}_- = \mathfrak{A}_+$ is satisfied. *Hence the pair* $(\mathfrak{A}; J)$ *is associator simple for* $r > 2$.

6. Finally we consider finitely many pairs $(\mathfrak{A}_1, J_1), \cdots, (\mathfrak{A}_q, J_q)$ of Jordan algebras $\mathfrak{A}_i$ with automorphisms J_i, $J_i^2 = \mathrm{Id}$. Then the direct sum $J = J_1 \oplus \cdots \oplus J_q$ is an automorphism of $\mathfrak{A} = \mathfrak{A}_1 \oplus \cdots \oplus \mathfrak{A}_q$ satisfying $J^2 = \mathrm{Id}$. The pair $(\mathfrak{A}; J)$ is called the direct sum of the pairs $(\mathfrak{A}_i, J_i)$ and we write

$$(\mathfrak{A}; J) = (\mathfrak{A}_1, J_1) \oplus \cdots \oplus (\mathfrak{A}_q, J_q).$$

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Conversely, consider a pair $(\mathfrak{A};J)$ and suppose that $\mathfrak{A}$ is semi simple and of finite dimension. Let $\mathfrak{B}$ be a simple ideal of $\mathfrak{A}$. Then $J\mathfrak{B}$ is also a simple ideal. In the case $J\mathfrak{B}=\mathfrak{B}$ put $\mathfrak{A}=\mathfrak{B}$ and in ths case $J\mathfrak{B}\neq\mathfrak{B}$ put $\mathfrak{A}_1=\mathfrak{B}\oplus J\mathfrak{B}$. Denote by J_1 the restriction of J on $\mathfrak{A}_1$. Then $(\mathfrak{A}_1,J_1)$ is a simple pair of type I resp. II and we have a direct sum decomposition $(\mathfrak{A};J)=(\mathfrak{A}_1,J_1)\oplus(\mathfrak{A}_0,J_0)$ where $\mathfrak{A}_0$ is a complimentary ideal of $\mathfrak{A}_1$ and J_0 is the restriction of J on $\mathfrak{A}_0$. Iterating this procedure we end up with a *decomposition of the pair* $(\mathfrak{A};J)$ *as a direct sum of simple pairs* $(\mathfrak{A}_1,J_1),\cdots,(\mathfrak{A}_q,J_q)$.

We use this result for the proof of

THEOREM 2. *Let* $\mathfrak{A}$ *be a finite dimensional semi simple Jordan algebra and let* J, $J^2=\mathrm{Id}$, *be an automorphism of* $\mathfrak{A}$. *Then*

(i) *there exists a non degenerate associative bilinear form* λ *of* $\mathfrak{A}$ *satisfying* $\lambda(Ja,b)=\lambda(a,Jb)$ *for* $a,b\in\mathfrak{A}$

(ii) *The subalgebra* $\mathfrak{A}_+$ *is semi simple.*

Proof. According to the above result we may restrict ourselves to a simple pair $(\mathfrak{A};J)$. We use the reduced trace RS (see [1], page 82) of the algebra $\mathfrak{A}$, which is a non-zero semi normal (see [1], page 33) linear form of $\mathfrak{A}$ with values in the algebraic closure of the ground field satisfying $RS(Ja\cdot b)=RS(a\cdot Jb)$ for all automorphism J of $\mathfrak{A}$ such that $J^2=\mathrm{Id}$ (see [1], Chapter III, Satz 5.2b and Corollary 1 of Satz 5.4 together with IV, Satz 6.1). Hence $\mathfrak{A}_+$ and $\mathfrak{A}_-$ are orthogonal with respect to RS and therefore the restriction of RS on $\mathfrak{A}_+$ has bilinear kernel 0. Since this restriction is again a semi normal linear form of $\mathfrak{A}_+$ we see that $\mathfrak{A}_+$ has radical 0 (see [1], Chapter I, § 7) and $\mathfrak{A}_+$ turns out to be semi simple.

The same arguments as in the proof of Lemma 6.1 and Satz 6.2 in [1], Chapter I, show that for a simple Jordan algebra $\mathfrak{A}$ there exists a non degenerate associative bilinear form $\mu_{\mathfrak{A}}$ with values in the ground field satisfying $\mu_{\mathfrak{A}}(Ja,b)=\mu_{\mathfrak{A}}(a,Jb)$. If $(\mathfrak{A};J)$ is of type II then we put $\lambda=\mu_{\mathfrak{A}}$. If $(\mathfrak{A};J)$ is of type I then we may assume $\mathfrak{A}=\mathfrak{B}\oplus\mathfrak{B}$ where $\mathfrak{B}$ is simple and where J is the canonical automorphism. In this case we put $\lambda=\mu_{\mathfrak{B}}\oplus\mu_{\mathfrak{B}}$.

3. On decomposing automorphisms.

1. Let $\mathfrak{A}$ be a Jordan algebra with unit element e and let J, $J^2=\mathrm{Id}$, be an automorphism of $\mathfrak{A}$. We consider the associator ideal $\mathfrak{c}=\mathfrak{c}(\mathfrak{A};J)$ of the pair $(\mathfrak{A};J)$ consisting of the elements $g\in\mathfrak{Z}(\mathfrak{A}_+)$ such that $g\mathfrak{A}_-=0$ (see the remark in § 1.3). As a consequence we obtain $\mathfrak{c}^2\subset\mathfrak{Z}(\mathfrak{A})$. Indeed,

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for $g \in \mathfrak{c}$ and $a, b \in \mathfrak{A}$ it follows $(a, b, g^2) = 2(ag, b, g) = 2(a_+g, b_+, g) = 0$ from (A') and $(a, g^2, b) = 2g(a, g, b) = 2g(a, g, b)_+ = 2g(a_+, g, b_+) = 0$ from (B').

We know from § 2.**4** that

$$\mathfrak{A}'' = \mathfrak{A}_-\mathfrak{A}_- + \mathfrak{A}_-$$

is an associator ideal of $(\mathfrak{A}; J)$ too. The automorphism J is called *decomposing for* $\mathfrak{A}$ if both

$$\mathfrak{A} = \mathfrak{c} + \mathfrak{A}'' \text{ and } \mathfrak{Z}(\mathfrak{A}_+) = \mathfrak{c} + \mathfrak{Z}(\mathfrak{A}) \cap \mathfrak{A}_+ \tag{3.1}$$

holds. Since $\mathfrak{c}$ is contained in $\mathfrak{A}_+$ we have the equivalent condition

$$\mathfrak{A}_+ = \mathfrak{c} + \mathfrak{A}_-\mathfrak{A}_- \text{ and } \mathfrak{Z}(\mathfrak{A}_+) = \mathfrak{c} + \mathfrak{Z}(\mathfrak{A}) \cap \mathfrak{A}_+ \tag{3.1'}$$

We call J *directly decomposing* if the first sum in (3.1) (or in (3.1')) is a direct sum decomposition of $\mathfrak{A}$ (resp. of $\mathfrak{A}_+$). The usual arguments show that J is directly decomposing for a finite dimensional algebra $\mathfrak{A}$ if and only if this is true for an arbitrary ground field extension $\tilde{\mathfrak{A}}$ of $\mathfrak{A}$. Furthermore, if $(\tilde{\mathfrak{A}}; J)$ is a direct sum of simple pairs $(\mathfrak{A}_i, J_i)$ then J is decomposing for $\mathfrak{A}$ if and only if the automorphisms J_i are decomposing for $\mathfrak{A}_i$. An analogous statement holds for directly decomposing automorphisms.

In particular, an automorphism $J \neq \mathrm{Id}$ is decomposing (even directly decomposing) whenever $(\mathfrak{A}; J)$ is an associator simple pair satisfying $\mathfrak{Z}(\mathfrak{A}_+) = \mathfrak{Z}(\mathfrak{A}) \cap \mathfrak{A}_+$ (because then $\mathfrak{c} = 0$ and $\mathfrak{A}'' = \mathfrak{A}$).

Using this necessary condition we conclude from § 2.**6** *that in the following two cases the automorphism* J *is decomposing for* $\mathfrak{A}$:

(A) $(\mathfrak{A}; J)$ is a simple pair of type I.

(B) $\mathfrak{A}$ is the Jordan algebra over K of $r \times r$ $(r > 2)$ hermitian matrices with entries in a quadratic extension $K(\vartheta)$ of K and J transposes the matrices.

In § 6 we shall give other examples of decomposing automorphisms. One verifies that $J = \mathrm{Id}$ is a decomposing automorphism if and only if the Jordan algebra is associative.

2. According to the remark in § 1.**4** we have for the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$

$$[\mathfrak{G}, \mathfrak{G}] = \mathfrak{h}^+ \oplus \mathfrak{A}', \tag{3.2}$$

where $\mathfrak{A}' = \mathfrak{h}^+\mathfrak{A}$. Since $[\mathfrak{G}, \mathfrak{G}]$ is an ideal in $\mathfrak{G}$ we conclude from Lemma 6 *that* $\mathfrak{A}'$ *is an associator ideal of* $(\mathfrak{A}; J)$. Furthermore from (1.1) it follows

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$$\mathfrak{A}' = \mathfrak{A}'' + (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_+). \tag{3.3}$$

Hence we have a chain $\mathfrak{c} \subset \mathfrak{A}'' \subset \mathfrak{A}' \subset \mathfrak{A}$ of associator ideals.

Using (I; 6.1) and (1.7) we conclude

(3.4) $\lambda(\mathfrak{c}, \mathfrak{A}') = 0$ whenever λ is an associative bilinear form of $\mathfrak{A}$ because $\lambda(\mathfrak{c}, \mathfrak{A}') = \lambda(\mathfrak{c}, \mathfrak{h}^+\mathfrak{A}) = \lambda(\mathfrak{h}^+\mathfrak{c}, \mathfrak{A}) = 0$. *Hence* $\mathfrak{c} \cap \mathfrak{A}' = 0$ *if* λ *is non degenerate. In particular, any decomposing automorphism* J *is directly decomposing whenever* $\mathfrak{A}$ *admits a non degenerate associative bilinear form.*

The center $\mathfrak{C} = \mathfrak{C}(\mathfrak{A}; J)$ of the Lie algebra $\mathfrak{G}$ has the form $\mathfrak{C} = 0 \oplus \mathfrak{c}$ (see § 1.**4**). Since (3.1) implies $\mathfrak{A} = \mathfrak{c} + \mathfrak{A}'$ we obtain from (3.2)

$$\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}] \tag{3.5}$$

whenever the automorphism J is decomposing.

LEMMA 8. *Let* $\mathfrak{A}$ *be a Jordan algebra admitting a non degenerate associative bilinear form. Let* J *be a decomposing automorphism of* $\mathfrak{A}$. *Then*

a) $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$ *is a direct sum composition,*

b) $\mathfrak{C} = \mathfrak{J}^+(\mathfrak{c})$.

Proof. a) From $\mathfrak{c} \cap \mathfrak{A}' = 0$ we conclude that the sum (3.5) is direct.

b) It suffices to prove $\mathfrak{j}^+(\mathfrak{c}) = 0$. For $T \in \mathfrak{j}^+(\mathfrak{c})$ we have $T \in \mathfrak{h}^+$ and $T\mathfrak{A} \subset \mathfrak{c}$. But using $T^* = -JTJ$, $J\mathfrak{c} \subset \mathfrak{c}$, we obtain also $T^*\mathfrak{A} \subset \mathfrak{c}$. Hence (3.4) leads to $\lambda(a, T\mathfrak{A}') = \lambda(T^*a, \mathfrak{A}') = 0$ and consequently $T\mathfrak{A}' = 0$. By the definition of $\mathfrak{c}$ we have $T\mathfrak{c} = 0$ and therefore $T = 0$.

4. Associative bilinear forms.

1. In this section we consider again a pair $(\mathfrak{A}; J)$, where $\mathfrak{A}$ is a Jordan algebra with unit element e. Let σ be a symmetric bilinear form of $\mathfrak{A}$ (with values in an extension field of the ground field). We are going to discuss some conditions for σ in connection with the associativity of certain bilinear forms of the Lie algebras $\mathfrak{h}^+$ and $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$. The bilinear form σ is called an *associative bilinear form of the pair* $(\mathfrak{A}; J)$ or simply *associative for* $(\mathfrak{A}; J)$ if

$$\sigma(Ja, b) = \sigma(a, Jb), \tag{i}$$

$$\sigma((a \nabla b)c, Jd) = \sigma((c \nabla d)a, Jb), \tag{ii}$$

is satisfied for $a, b, c, d \in \mathfrak{A}$. In particular from (i) it follows that $\mathfrak{A}_+$ and $\mathfrak{A}_-$ are orthogonal with respect to σ. Since $\mathfrak{h}^+$ is spanned by elements of the

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form $c \nabla d$ and since the left hand side of (ii) changes the sign by interchanging a and b we obtain

$$\sigma(Ta, Jb) + \sigma(Tb, Ja) = 0 \text{ for } T \in \mathfrak{h}^+. \tag{4.1}$$

Substituting $T = L(u)$, $u \in \mathfrak{A}_-$, and replacing b by Jb we observe

$$\sigma(au, b) = \sigma(a, ub), \quad u \in \mathfrak{A}_-. \tag{4.2}$$

We prove *that the bilinear kernel* $\mathfrak{m} = Bk_\sigma(\mathfrak{A})$ *is an associator ideal of* $(\mathfrak{A}; J)$ *whenever* σ *is an associative bilinear form of the pair* $(\mathfrak{A}; J)$. Indeed, $\mathfrak{h}^+\mathfrak{m} \subset \mathfrak{m}$ follows from (4.1) while $(\mathfrak{A} \nabla \mathfrak{m})\mathfrak{A} \subset \mathfrak{m}$ is a consequence of (ii).

Let λ be an associative bilinear form of the algebra $\mathfrak{A}$. Then one verifies $\lambda((a \square b)c, d) = \lambda((c \square d)a, b)$ for $a, b, c, d \in \mathfrak{A}$. Using $J(c \square d)J = Jc \square Jd$ we conclude *that any associative bilinear form of* $\mathfrak{A}$ *satisfying* (i) *is associative for the pair* $(\mathfrak{A}; J)$. But conversely it is *not* true that the associativity of σ is a consequence of (i) and (ii).

2. Let σ be an associative bilinear form of the pair $(\mathfrak{A}; J)$. Since $\mathfrak{h}^+$ is spanned by transformations of the form $a \nabla b$ we may define a bilinear form $\sigma_{\mathfrak{h}^+}$ of $\mathfrak{h}^+$ by[1]

$$\sigma_{\mathfrak{h}^+}(T, S) = 4\Sigma\sigma(Tc, Jd) = 4\Sigma\sigma(Sa, Jb), \tag{4.3}$$

whenevr

$$T = \Sigma(a \nabla b) \in \mathfrak{h}^+, \quad S = \Sigma(c \nabla d) \in \mathfrak{h}^+.$$

From (ii) it follows that in this definition the second and the third terms coincide. Hence $\sigma_{\mathfrak{h}^+}$ is well-defined and symmetric on $\mathfrak{h}^+$. In order to prove *that* $\sigma_{\mathfrak{h}^+}$ *is an associative bilinear form of* $\mathfrak{h}^+$ *whenever* σ *is associative for* $(\mathfrak{A}; J)$ we use Lemma 3 and (4.1) and we get

$$\begin{aligned}
&\sigma_{\mathfrak{h}^+}([T_1, a \nabla b], T_2) \\
&\quad = \sigma_{\mathfrak{h}^+}(T_1 a \nabla b + a \nabla T_1 b, T_2) = 4\sigma(T_2 T_1 a, Jb) + 4\sigma(T_2 a, JT_1 b) \\
&\quad = 4\sigma([T_2, T_1]a, Jb).
\end{aligned}$$

Therefore the left hand side changes the sign by interchanging 1 and 2.

Next we define a bilinear form $\sigma_{\mathfrak{G}}$ for the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ by setting

$$\sigma_{\mathfrak{G}}(x_1, x_2) = \sigma_{\mathfrak{h}^+}(T_1, T_2) + 4\sigma(a_1, Ja_2) \tag{4.4}$$

whenever $x_i = T_i \oplus a_i \in \mathfrak{G}$. The bilinear form $\sigma_{\mathfrak{G}}$ is an unique extension of σ in the following sense:

[1] The factor 4 brings a simplification in 4.

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LEMMA 9. *Let σ be a bilinear form of $\mathfrak{A}$ satisfying $\sigma(Ja,b)=\sigma(a,Jb)$. Then the following two statements are equivalent:*

a) *There exists a bilinear form σ' of $\mathfrak{h}^+$ such that the bilinear form σ^* given by $\sigma^*(x_1,x_2)=\sigma'(T_1,T_2)+4\sigma(a_1,Ja_2)$, $x_i=T_i\oplus a_i$, of $\mathfrak{G}$ is associative.*

b) *σ is associative for the pair $(\mathfrak{A};J)$.*

COROLLARY. *$\sigma_{\mathfrak{G}}$ is associative for $\mathfrak{G}$ if and only if σ is associative for the pair $(\mathfrak{A};J)$. If this is true we have $\sigma'=\sigma_{\mathfrak{h}^+}$ and $\sigma^*=\sigma_{\mathfrak{G}}$.*

Proof. One verifies that σ^* is associative if and only if σ' is associative as well as

$$\sigma'(a\nabla b,T)=4\sigma(Ta,Jb)=-4\sigma(Tb,Ja) \tag{4.5}$$

holds for $a,b\in\mathfrak{A}$ and $T\in\mathfrak{h}^+$.

Suppose first that σ^* is associative. Then the substitution $T=c\nabla d$ in (4.5) leads to (ii) because σ' is symmetric. Furthermore (4.5) yields $\sigma'=\sigma_{\mathfrak{h}^+}$.

Conversely let σ be associative for the pair $(\mathfrak{A};J)$. Then (4.5) is satisfied for the associative bilinear form $\sigma'=\sigma_{\mathfrak{h}^+}$ because of (4.3).

Let σ be an associative bilinear form of the pair $(\mathfrak{A};J)$. We know that the bilinear kernel $Bk_\sigma(\mathfrak{A})$ is an associator ideal of $(\mathfrak{A};J)$. In order to prove

$$Bk_{\sigma_{\mathfrak{G}}}(\mathfrak{G})=\mathfrak{F}^+(Bk_\sigma(\mathfrak{A})) \tag{4.6}$$

We conclude from (4.4) that the left hand side of (4.6), i.e. the bilinear kernel of $\mathfrak{G}$ with respect to $\sigma_{\mathfrak{G}}$ coincides with $\mathfrak{g}\oplus\mathfrak{m}$ where $\mathfrak{g}$ denotes the bilinear kernel of $\mathfrak{h}^+$ with respect to $\sigma_{\mathfrak{h}^+}$ and where $\mathfrak{m}=Bk_\sigma(\mathfrak{A})$. Using (4.3) and $\mathfrak{h}^+=\mathfrak{A}\nabla\mathfrak{A}$ we see that $T\in\mathfrak{g}$ is equivalent to $\sigma(Ta,Jb)=0$ for $a,b\in\mathfrak{A}$, hence to $T\mathfrak{A}\subset\mathfrak{m}$. Now (4.6) follows from the definition of $\mathfrak{F}^+(\mathfrak{m})$ (see § 2. **2**).

3. We apply our construction to an associative bilinear form σ of $\mathfrak{A}$ satisfying (i). In I, § 6. **6**, we gave a construction of associative bilinear form $\sigma_{\mathfrak{h}}$ resp. $\sigma_{\mathfrak{L}}$ of the Lie algebras $\mathfrak{h}=\mathfrak{h}(\mathfrak{A})$ resp. $\mathfrak{L}=\mathfrak{L}(\mathfrak{A})$. From I, Lemma 12, it follows $\sigma_{\mathfrak{h}}(a\,\square\,b,T)=2\sigma(Ta,b)$ for $a,b\in\mathfrak{A}$ and $T\in\mathfrak{h}$. A combination with (4.3) shows *that $\sigma_{\mathfrak{h}^+}$ coinsides with the restriction of $\sigma_{\mathfrak{h}}$ on $\mathfrak{h}^+$.*

Using the isomorphism π of $\mathfrak{G}$ into $\mathfrak{L}$ (see Lemma 2) one verifies $\sigma_{\mathfrak{L}}\circ\pi=\sigma_{\mathfrak{G}}$.

Combining Theorem 2 and (4.6) we end up with

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THEOREM 3. *Let $\mathfrak{A}$ be a finite dimensional semi simple Jordan algebra and let J, $J^2 = \mathrm{Id}$, be an automorphism of $\mathfrak{A}$. Then there exists an associative bilinear form σ of $\mathfrak{A}$ such that $\sigma_{\mathfrak{G}}$ is a non degenerate associative bilinear form of $\mathfrak{G}$.*

4. We restrict ourselves again to a finite dimensional semi simple Jordan algebra $\mathfrak{A}$. According to Theorem 2 let λ be a non degenerate associative bilinear form of $\mathfrak{A}$ satisfying (i). For a linear transformation F of $\mathfrak{A}$ denote by F^* the adjoint transformation with respesct to λ. Note that the adjoint of $T \in \mathfrak{h}^+$ coincides with the image of T under the antiautomorphism $T \to T^*$ of $\mathfrak{h}^+$ (see I, formula (6.1)).

In order to describe all associative bilinear forms σ of the pair $(\mathfrak{A}; J)$ with values in the ground field we write for a given one

$$\sigma(a, b) = \lambda(Fa, b), \quad F = F^* \in \mathrm{Hom}(\mathfrak{A}, \mathfrak{A}).$$

PROPOSITION. *Let be $F = F^* \in \mathrm{Hom}(\mathfrak{A}, \mathfrak{A})$. Then $\sigma(a, b) = \lambda(Fa, b)$ is an asociative bilinear form for $(\mathfrak{A}; J)$ if and only if*

(1) $JF = FJ$,

(2) $F(a, u, b) = (Fa, u, b) = (a, Fu, b) = (a, u, Fb)$ *for* $u \in \mathfrak{A}$ *and* $a, b \in \mathfrak{A}_+$ *or* $a, b \in \mathfrak{A}_-$.

(3) $F(au) = a \cdot Fu = u \cdot Fa$ *for* $a \in \mathfrak{A}_-$ *and* $u \in \mathfrak{A}$.

Proof. First of all (i) and (1) are equivalent. We prove that (ii) is equivalent to

(ii′) $F(a \nabla b) = Fa \nabla b = a \nabla Fb = (a \nabla b)F$ for $a, b \in \mathfrak{A}$.

Indeed, (ii) is equivalent to $\lambda(F(a \nabla b)c, Jd) = \lambda((c \nabla d)a, JFb)$ because of (1). Since λ is also associative for $(\mathfrak{A}; J)$ (see **1**) we get $\lambda(F(a \nabla b)c, Jd) = \lambda((a \nabla Fb)c, Jd)$ and hence $F(a \nabla b) = a \nabla Fb$ as equivalent condition to (ii). But $a \nabla b = -b \nabla a$ and an application of the antiautomorphism $T \to T^*$ on $F(a \nabla b) = Fa \nabla b$ leads to (ii′).

From (2.1) it follows that (ii′) is equivalent to

(4.7) $FL(ab) = L(Fa \cdot b) = L(a \cdot Fb) = L(ab)F$, $a \in \mathfrak{A}_-$, $b \in \mathfrak{A}_+$,

together with

$$F[L(a), L(b)] = [L(Fa), L(b)] = [L(a), L(Fb)] = [L(a), L(b)]F$$

for $a, b \in \mathfrak{A}_+$ or $a, b \in \mathfrak{A}_-$, i. e. with (2). Substituting $b = e$ in (4.7) we obtain

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(3). But conversely we get $FL(a) = L(Fa)$, $a \in \mathfrak{A}_-$, from (3) and therefore (4.7) is implied by (3).

From (1) it follows $F\mathfrak{A}_+ \subset \mathfrak{A}_+$ and $F\mathfrak{A}_- \subset \mathfrak{A}_-$. Put

$$f = Fe \in \mathfrak{A}_+.$$

Then (2) implies

$$(2') \qquad f \in \mathfrak{Z}(\mathfrak{A}_+) \text{ and } (f, \mathfrak{A}, \mathfrak{A}_+) = 0,$$

and (3) leads to

$$(3') \qquad Fa = fa \ (a \in \mathfrak{A}_-) \text{ and } F(ab) = a(fb) = b(fa) \ (a, b \in \mathfrak{A}_-).$$

We conclude for the associator ideals $\mathfrak{c}$ and $\mathfrak{A}'' = \mathfrak{A}_-\mathfrak{A}_- + \mathfrak{A}_-$ from (2) and (3) that $F\mathfrak{c} \subset \mathfrak{c}$ and from (3') that $F\mathfrak{A}'' \subset \mathfrak{A}''$ holds.

Denote by σ'' the restriction of σ to $\mathfrak{A}''$. We obtain from (3')

$$(4.8) \qquad \begin{aligned} &\sigma''(a, b) = \lambda(fa, b), \ \sigma''(a, bc) = \sigma''(bc, a) = 0, \\ &\sigma''(ab, cd) = \lambda(a \cdot fb, cd) = \lambda(ab, c \cdot fd) \end{aligned}$$

for $a, b, c, d \in \mathfrak{A}_-$. *Hence σ'' depends only on $f = Fe$.* But f can be replaced by $f + g$ where g is an arbitrary element of $\mathfrak{c}$.

5. Finally we suppose that J is a decomposing automorphism. Hence J is directly decomposing, in particular $\mathfrak{A} = \mathfrak{c} + \mathfrak{A}''$ is a direct sum decomposition. According to this decomposition we write $a = a_0 + a''$ for the elements a of $\mathfrak{A}$. From (3.1) it follows that there exists an element $g \in \mathfrak{c}$ such that $f + g$ belongs to the center $\mathfrak{Z}(\mathfrak{A})$ of $\mathfrak{A}$. Hence we may assume that $f \in \mathfrak{Z}(\mathfrak{A})$, then from (4.8) we get $\sigma''(a'', b'') = \lambda(fa'', b'')$.

We denote the restriction of F on $\mathfrak{c}$ by F_0. Hence any associative bilinear form σ of the pair $(\mathfrak{A}; J)$ can be written as

$$(4.9) \qquad \sigma(a, b) = \lambda(fa'', b'') + \lambda(F_0 a_0, b_0) \text{ where } f \in \mathfrak{Z}(\mathfrak{A}) \text{ and } F_0 \in \mathrm{Hom}(\mathfrak{c}, \mathfrak{c}).$$

As an application we prove

THEOREM 4. *Let $\mathfrak{A}$ be a finite dimensional semi simple Jordan algebra and let J, $J^2 = \mathrm{Id}$, be a decomposing automorphism of $\mathfrak{A}$. Moreover let λ be a non degenerate associative bilinear form of $\mathfrak{A}$ satisfying $\lambda(Ja, b) = \lambda(a, Jb)$.*

Then to any associative bilinear form σ of the pair $(\mathfrak{A}; J)$ satisfying $\mathfrak{c} \subset Bk_\sigma(\mathfrak{A})$ there exists an element f in the center $\mathfrak{Z}(\mathfrak{A})$ of $\mathfrak{A}$ such that $\sigma(a, b) = \lambda(fa'', b'')$, where a'' resp. b'' are the components of a resp. b in $\mathfrak{A}''$.

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Proof. Using (4.9) and the fact that $\mathfrak{c}$ and $\mathfrak{A}''$ are orthogonal with respect to λ (see § 3.2) we see that $\mathfrak{c} \subset Bk_\sigma(\mathfrak{A})$ is equivalent to $F_0 = 0$.

5. The decompositions $\mathfrak{G} = \mathfrak{K}_\epsilon + \mathfrak{P}_\epsilon$.

1. At the beginning we consider a pair $(\mathfrak{A}; J)$ where $\mathfrak{A}$ is only an admissible algebra with unit element e. The automorphisms Θ_ϵ, $\epsilon = 0, \pm 1$, of $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ (see § 1.4) given by

$$\Theta_0 x = T \oplus (-a), \quad \Theta_\epsilon x = (-T^*) \oplus \epsilon Ja \quad (\epsilon = \pm 1),$$

where $x = T \oplus a \in \mathfrak{G}$ induce direct sum decompositions

$$\mathfrak{G} = \mathfrak{K}_\epsilon + \mathfrak{P}_\epsilon \quad (\epsilon = 0, \pm 1),$$

where

$$\mathfrak{K}_\epsilon = \{x; x \in \mathfrak{G}, \Theta_\epsilon x = x\}, \quad \mathfrak{P}_\epsilon = \{x; x \in \mathfrak{G}, \Theta_\epsilon x = -x\}.$$

Obviously we have

$$[\mathfrak{K}_\epsilon, \mathfrak{K}_\epsilon] \subset \mathfrak{K}_\epsilon, [\mathfrak{K}_\epsilon, \mathfrak{P}_\epsilon] \subset \mathfrak{P}_\epsilon, \ [\mathfrak{P}_\epsilon, \mathfrak{P}_\epsilon] \subset \mathfrak{K}_\epsilon.$$

Using the notation

$$\mathfrak{d}^\eta = \{T; T \in \mathfrak{h}^+, -T^* = \eta T\}, \quad (\eta = \pm 1)$$

we obtain a direct sum decomposition $\mathfrak{h}^+ = \mathfrak{d}^+ + \mathfrak{d}^-$ satisfying the analogous compositions rules. One verifies

$$\mathfrak{K}_0 = \mathfrak{h}^+ = \mathfrak{d}^+ + \mathfrak{d}^-, \quad \mathfrak{K}_\epsilon = \mathfrak{d}^+ \oplus \mathfrak{A}_\epsilon,$$
$$\mathfrak{P}_0 = \mathfrak{A} = \mathfrak{A}_+ + \mathfrak{A}_-, \quad \mathfrak{P}_\epsilon = \mathfrak{d}^- \oplus \mathfrak{A}_{-\epsilon},$$

for $\epsilon = \pm 1$. We write the indices $\epsilon = \pm 1$ as "$+$" resp. "$-$." In an obvious manner we write $T = T^+ + T^-$ for $T \in \mathfrak{h}^+$ and $a = a_+ + a_-$ for $a \in \mathfrak{A}$. Furthermore we write for $x = T \oplus a \in \mathfrak{G}$:

$$x_\epsilon{}^\eta = T^\eta \oplus a_{\epsilon\eta}, \quad \epsilon, \eta = \pm 1,$$

and we obtain

$$\mathfrak{K}_\epsilon = \{x_\epsilon{}^+; x \in \mathfrak{G}\}, \quad \mathfrak{P}_\epsilon = \{x_\epsilon{}^-; x \in \mathfrak{G}\} \quad (\epsilon = \pm 1)$$

as well as $x = x_\epsilon{}^+ + x_\epsilon{}^-$ for $\epsilon = \pm 1$.

The automorphisms Θ_0, Θ_+ and Θ_- map each space $\mathfrak{K}_\epsilon$ resp. $\mathfrak{P}_\epsilon$ ($\epsilon = 0, \pm 1$) into itself because they commute mutually. Since $\mathfrak{c}$ is contained in $\mathfrak{A}_+$ we obtain $\mathfrak{C} = 0 \oplus \mathfrak{c} \subset \mathfrak{K}_+ \cap \mathfrak{P}_-$.

Remark. For a Jordan algebra $\mathfrak{A}$ we have

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$$\mathfrak{d}^+ = [L(\mathfrak{A}_+), L(\mathfrak{A}_+)] + [L(\mathfrak{A}_-), L(\mathfrak{A}_-)], \quad \mathfrak{d}^- = L(\mathfrak{A}_-).$$

In this case there is a further decomposition of $\mathfrak{P}_+$. Using the subspaces $\mathfrak{p}^+$ resp. $\mathfrak{p}^-$ given by

$$\mathfrak{p}^\eta = \{2L(w) \oplus \eta w ; w \in \mathfrak{A}\} \quad (\eta = \pm 1)$$

we obtain $\mathfrak{P}_+ = \mathfrak{p}^+ + \mathfrak{p}^-$. *Both spaces* $\mathfrak{p}^+$ *and* $\mathfrak{p}^-$ *are abelian subalgebras of* $\mathfrak{G}$. Indeed, for $x_i = 2L(w_i) \oplus \eta w_i \in \mathfrak{p}^\eta$ we have $[x_1, x_2] = T \oplus 0$ where

$$T = 4[L(w_1), L(w_2)] + w_1 \nabla w_2 = 0.$$

Moreover one verifies $\Theta_- \mathfrak{p}^\eta = \mathfrak{p}^{-\eta}$ for $\eta = \pm 1$.

2. We restrict ourselves to finite dimensional Jordan algebras $\mathfrak{A}$. For a linear transformation T of $\mathfrak{A}$ which maps $\mathfrak{A}_+$ and $\mathfrak{A}_-$ into itself, we denote by $\mathrm{Trace}_+ T$ resp. $\mathrm{Trace}_- T$ the trace of the restriction of T to $\mathfrak{A}_+$ resp. $\mathfrak{A}_-$.

We introduce a symmetric bilinear form $\delta = \delta_J$ of $\mathfrak{A}$ by setting

$$\delta(a, b) = \mathrm{Trace}\{L(ab) - P(a, Jb)J\},$$

where $P(a, b)$ stands for the linearisation of $P(a) = 2L^2(a) - L(a^2)$. Since J is an automorphism of $\mathfrak{A}$ we conclude $\delta(a, Jb) = \delta(b, Ja)$ for $a, b \in \mathfrak{A}$. Hence $\mathfrak{A}_+$ and $\mathfrak{A}_-$ are orthogonal with respect to δ.

The Killing form of a Lie algebra $\mathfrak{q}$ is denoted by $\langle\ ,\ \rangle_\mathfrak{q}$. Using the three decompositions $\mathfrak{G} = \mathfrak{K}_\epsilon + \mathfrak{P}_\epsilon$, $\epsilon = 0, \pm 1$, we are going to calculate the Killing forms of the algebras $\mathfrak{G}$ and $\mathfrak{K}_\epsilon$. The tool for it will be the following

PROPOSITION. *Let* $\mathfrak{k} + \mathfrak{p}$ *be a direct sum composition of a Lie algebra satisfying* $[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}$, $[\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}$, $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$. *Then*

$$\langle k + p, k + p\rangle_{\mathfrak{k}+\mathfrak{p}} = \langle k, k\rangle_\mathfrak{k} + t_\mathfrak{p}(k, p), \quad k + p \in \mathfrak{k} + \mathfrak{p},$$

where $t_\mathfrak{p}(k, p) = \mathrm{Trace}_\mathfrak{p} M(k) + 2\,\mathrm{Trace}_\mathfrak{p} M(p)$ *and where* $M(y)z = [y, [y, z]]$. For a proof see [1], Chapter IX, Lemma 5.2.

LEMMA 10. *For* $x = T \oplus a \in \mathfrak{G}$, $T^+ = D \in \mathfrak{d}^+$, $T^- = L(d)$, $d \in \mathfrak{A}_-$, *one has*

(a) $\langle x, x\rangle_\mathfrak{G} = \langle T, T\rangle_{\mathfrak{d}^+} + \mathrm{Trace}\, T^2 + 4\delta(a, Ja)$,

(b) $\langle x, x\rangle_\mathfrak{G} = \langle x_\epsilon^+, x_\epsilon^+\rangle_{\mathfrak{K}_\epsilon} + \mathrm{Trace}_-\{D^2 + 2L(d^2)\} + \mathrm{Trace}_{-\epsilon} D^2$
$\quad + 4\delta(a, Ja) - 8\epsilon\,\mathrm{Trace}_\epsilon\{L(a_\epsilon^2) - L^2(a_\epsilon)\}$,

(c) $\langle T, T\rangle_{\mathfrak{d}^+} = \langle D, D\rangle_{\mathfrak{d}^+} + \mathrm{Trace}_-\{D^2 + 2L(d^2) - 2L^2(d)\}$,

(d) $\langle x_\epsilon^+, x_\epsilon^+\rangle_{\mathfrak{K}_\epsilon} = \langle D, D\rangle_{\mathfrak{d}^+} + \mathrm{Trace}_\epsilon\{D^2 + 8\epsilon[L(a_\epsilon^2) - L^2(a_\epsilon)]\}$.

Moreover in the proof we shall obtain

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$$(5.1) \qquad \mathrm{Trace}_+ L^2(d) = \mathrm{Trace}_- L^2(d) = \tfrac{1}{2}\mathrm{Trace}\, L^2(d), \quad d \in \mathfrak{A}_-.$$

Proof. In order to prove (a), (c) and (d) by applying the proposition to the decomposition $\mathfrak{G} = \mathfrak{h}^+ \oplus \mathfrak{A}$, $\mathfrak{h}^+ = \mathfrak{d}^+ + L(\mathfrak{A}_-)$ and $\mathfrak{K}_\epsilon = \mathfrak{d}^+ \oplus \mathfrak{A}_\epsilon$ $(\epsilon = \pm 1)$ it suffices to show

(a′) $t(T, a) = \mathrm{Trace}\, T^2 + 4\delta(a, Ja)$,

(c′) $t(D, L(d)) = \mathrm{Trace}_-\{D^2 + 2L(d^2) - 2L^2(d)\}$,

(d′) $t(D, a_\epsilon) = \mathrm{Trace}_\epsilon\{D^2 + 8\epsilon[L(a_\epsilon^2) - L^2(a_\epsilon)]\}$.

From $M(T)b = [T, [T, b]] = T^2 b$ and from

$$M(a)b = [a, [a, b]] = [a, a \nabla b]$$
$$= (b \nabla a)a = (b \square Ja)a - (a \square Jb)a = (a \square Ja)b - 2P(a)Jb$$

we obtain (a′).

From $M(D)L(f) = [D, [D, L(f)]] = L(D^2 f)$, $f \in \mathfrak{A}_-$, and from

$M(L(d))L(f) = [L(d), [L(d), L(f)]] = L(d^2 f - d(df))$ *it* follows (c′).

Finally for $f \in \mathfrak{A}_\epsilon$ we have $M(D)f = D^2 f$ and

$$M(a_\epsilon)f = [a_\epsilon, a_\epsilon \nabla f] = (f \nabla a_\epsilon)a_\epsilon$$
$$= \epsilon(f \square a_\epsilon)a_\epsilon - \epsilon(a_\epsilon \square f)a_\epsilon = 4\epsilon[L(f), L(a_\epsilon)]a_\epsilon = 4\epsilon[L(a_\epsilon^2) - L^2(a_\epsilon)]f$$

and we observe (d′).

Next we have to prove formula (5.1). For $p = L(d)$ we apply the proposition to $\mathfrak{k} = \mathfrak{K}_\epsilon$, $\mathfrak{p} = \mathfrak{P}_\epsilon$ $(\epsilon = \pm 1)$ and we obtain $\frac{1}{2}\langle L(d), L(d)\rangle_{\mathfrak{G}}$ $= \mathrm{Trace}_{\mathfrak{P}_\epsilon} M(L(d))$. Let be $f \in \mathfrak{A}_-$ and $g \in \mathfrak{A}_{-\epsilon}$. Then it follows

$$M(L(d))(L(f) \oplus g)$$
$$= [L(d), [L(d), L(f)]] \oplus d(dg) = L(d^2 f - d(df)) \oplus d(dg).$$

Hence

$$\tfrac{1}{2}\langle L(d), L(d)\rangle_{\mathfrak{G}} - \mathrm{Trace}_-[L(d^2) - L^2(d)] = \mathrm{Trace}_{-\epsilon} L^2(d).$$

Since the left hand side does not depend on ϵ we end up with formula (5.1).

For the completion of the proof we use again the proposition for $\mathfrak{G} = \mathfrak{h}^+ \oplus \mathfrak{A} = \mathfrak{K}_\epsilon + \mathfrak{P}_\epsilon$ and we get

$$\langle T, T\rangle_{\mathfrak{h}^+} + t(T, a) = \langle x_\epsilon^+, x_\epsilon^+\rangle_{\mathfrak{K}_\epsilon} + t(x_\epsilon^+, x_\epsilon^-).$$

Hence

$$t(x_\epsilon^+, x_\epsilon^-) = t(T, a) + t(D, L(d)) - t(D, a_\epsilon)$$

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and substituting the formulas (a′), (c′), (d′) using (5.1) we end up with (b).

A combination of Lemma 9 and Lemma 10(a) leads to

COROLLARY 1. *The bilinear form* δ *is associative for the pair* $(\mathfrak{A};J)$ *and the Killing form of* $\mathfrak{G}$ *coincides with* $\delta_{\mathfrak{G}}$.

Since the center $\mathfrak{C}=0\oplus\mathfrak{c}$ of the Lie algebra $\mathfrak{G}$ is contained in the bilinear kernel of the Killing form of $\mathfrak{G}$, we conclude from (4.6) the

COROLLARY 2. $\mathfrak{c}\subset Bk_{\delta}(\mathfrak{A})$.

3. Suppose that $\mathfrak{A}$ admits a non degenerate associative bilinear form λ satisfying $\lambda(Ja,b)=\lambda(a,Jb)$ and suppose that J is a decomposing automorphism of $\mathfrak{A}$. Hence J is directly decomposing (see § 3.**2**). Combining Corollary 2 and Theorem 4 we obtain

$$\delta(a,b)=\lambda(fa'',b'') \text{ where } f\in\mathfrak{Z}(\mathfrak{A}). \tag{5.2}$$

Suppose that the element f is invertible. Since $\mathfrak{c}$ and $\mathfrak{A}''$ are orthogonal with respect to λ we get $Bk_{\delta}(\mathfrak{A})=\{a;a''=0\}=\mathfrak{c}$. An application of (4.6) and of Corollary 1 shows that *the bilinear kernel of the Killing form of* $\mathfrak{G}$ *coincides with* $\mathfrak{Z}^{+}(\mathfrak{c})$ and consequently *with the cener* $\mathfrak{C}$ *of* $\mathfrak{G}$ (see Lemma 8). We use this result for the proof of

THEOREM 5. *Let* $\mathfrak{A}$ *be a finite dimensional semi simple Jordan algebra over a field of characteristic* 0 *and let* J *be a decomposing automorphism of* $\mathfrak{A}$. *Then*

a) *the associative bilinear form* δ *of the pair* $(\mathfrak{A};J)$ *given by* $\delta(a,b)=\operatorname{Trace}[L(ab)-P(a,Jb)J]$ *has the bilinear kernel* $\mathfrak{c}$.

b) *The center* $\mathfrak{C}$ *of the Lie algebra* $\mathfrak{G}$ *coincides with the bilinear kernel of the Killing form* $\delta_{\mathfrak{G}}$ *of* $\mathfrak{G}$.

Proof. The pair $(\mathfrak{A};J)$ can be written as a direct sum of simple pairs $(\mathfrak{A}_i,J_i)$, where the J_i are decomposing automorphism of $\mathfrak{A}_i$. Hence $\mathfrak{G}=\mathfrak{G}(\mathfrak{A};J)$ turns out to be the direct sum of the Lie algebra $\mathfrak{G}(\mathfrak{A}_i,J_i)$ and it suffices to prove the theorem for a simple pair $(\mathfrak{A};J)$. It is known that the bilinear form given by $\operatorname{Trace}L(ab)$ satisfies all conditions for λ. We have to discuss the following three cases (see § 2.**5**):

a) The pair $(\mathfrak{A};J)$ is of type I: Hence we may assume that $\mathfrak{A}=\mathfrak{B}\oplus\mathfrak{B}$ and that J is the canonical automorphism of $\mathfrak{B}\oplus\mathfrak{B}$. One verifies $\operatorname{Trace}P(a,Jb)J=0$ for $a,b\in\mathfrak{A}$. Hence $f=e$ is invertible and we get the result from the above considerations.

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b) The pair $(\mathfrak{A}; J)$ is of type II and $J \neq \mathrm{Id}$: By the same argument as in a) it suffices to show that f is invertible. Since $f \in \mathfrak{Z}(\mathfrak{A})$ and since the center $\mathfrak{Z}(\mathfrak{A})$ is a field we would have $f = 0$ otherwise, i.e. $\delta(a, b) = 0$. But $\delta(e, e) = \mathrm{Trace}(\mathrm{Id} - J) = \dim \mathfrak{A}_- \neq 0$.

c) The pair $(\mathfrak{A}; J)$ is of type II and $J = \mathrm{Id}$: We know from § 3.1 that then $\mathfrak{A}$ is associative. Hence $\delta = 0$, $\mathfrak{h}^+ = 0$ and $\mathfrak{G} = 0 \oplus \mathfrak{A} = 0 \oplus \mathfrak{c} = \mathfrak{C}$.

COROLLARY 1. *The Killing form of the Lie algebra* $[\mathfrak{G}, \mathfrak{G}]$ *is non degenerate.*

Proof. From Lemma 8 we know that $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$ is a direct sum decomposition. Hence the Killing form of $[\mathfrak{G}, \mathfrak{G}]$ coincides with the restriction of the Killing form of $\mathfrak{G}$ on $[\mathfrak{G}, \mathfrak{G}]$.

COROLLARY 2. $\mathfrak{G}$ *is reductive, i.e. the decomposition* $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$ *is direct and* $[\mathfrak{G}, \mathfrak{G}]$ *is a direct sum of simple ideals or* 0.

Proof. The statement follows from Lemma 8 together with Corollary 1 and the criterion of Killing-Cartan.

6. On the automorphisms induced by idempotents.

1. Let $\mathfrak{A}$ be a Jordan algebra with unit element e of arbitrary dimension. In order to construct a class of automorphisms of $\mathfrak{A}$ we start with an idempotent $c \neq 0$ different from e. Let

$$\mathfrak{A} = \mathfrak{A}_{11} + \mathfrak{A}_{22} + \mathfrak{A}_{12}$$

be the Peirce decomposition of $\mathfrak{A}$ with respect to the orthogonal system $e_1 = c$, $e_2 = e - c$ of idempotents. Defining a bijection J_c of $\mathfrak{A}$ by $J_c = u_{11} + u_{22} - u_{12}$ whenever $u = u_{11} + u_{22} + u_{12}$, $u_{ij} \in \mathfrak{A}_{ij}$, we obtain an automorphism J_c of $\mathfrak{A}$ satisfying $J_c^2 = \mathrm{Id}$ and $\mathfrak{A}_+ = \mathfrak{A}_{11} + \mathfrak{A}_{22}$ resp. $\mathfrak{A}_- = \mathfrak{A}_{12}$. It is well known (and easy to see) that J_c coincides with $P(e_1 - e_2)$, where P denotes the quadratic representation of $\mathfrak{A}$.

Using two of the basic identities involving the Peirce spaces namely

$$a_i(b_i u_{ij}) + b_i(a_i u_{ij}) = (a_i b_i) u_{ij}, \quad a_i, b_i \in \mathfrak{A}_{ii}, u_{ij} \in \mathfrak{A}_{ij}, \tag{6.1}$$

$$e_i[u_{ij}(a_i v_{ij}) + v_{ij}(a_i u_{ij})] = a_i(u_{ij} v_{ij}), \quad a_i \in \mathfrak{A}_{ii}, u_{ij}, v_{ij} \in \mathfrak{A}_{ij} \tag{6.2}$$

(see for instance: formula (V.5) and (V.6) in [1], page 246) one concludes the following

PROPOSITION. *If* $\mathfrak{w}$ *is a subspace of* $\mathfrak{A}_{12}$ *which is invariant under* $\mathfrak{A}_{ii}$, *then both* $\mathfrak{a}_i(\mathfrak{w}) = \{x; x \in \mathfrak{A}_{ii}, x\mathfrak{A}_{12} \subset \mathfrak{w}\}$ *and* $\mathfrak{b}_i(\mathfrak{w}) = e_i(\mathfrak{A}_{12}\mathfrak{w})$ *are ideals of* $\mathfrak{A}_{ii}$.

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2. Again we use the abbreviation $\mathfrak{A}'' = \mathfrak{A}_{12} + \mathfrak{A}_{12}\mathfrak{A}_{12}$ resp.

$$\mathfrak{A}' = \mathfrak{A}'' + (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_+) = \mathfrak{A}'' + (\mathfrak{A}_{11}, \mathfrak{A}_{11}, \mathfrak{A}_{11}) + (\mathfrak{A}_{22}, \mathfrak{A}_{22}, \mathfrak{A}_{22})$$

and we know from § 2.**4** resp. § 3.**2** that $\mathfrak{A}'$ and $\mathfrak{A}''$ are associator ideals of $(\mathfrak{A}; J_c)$.

LEMMA 11. *If $\mathfrak{A}$ and $\mathfrak{A}_{ii}$ are simple then $\mathfrak{A}'' = \mathfrak{A}'$.*

Proof. From the proposition it follows that $e_i(\mathfrak{A}_{12}\mathfrak{A}_{12}) = \mathfrak{A}_{ii}$ because otherwise $\mathfrak{A}_{12}\mathfrak{A}_{12} \subset \mathfrak{A}_{ii}$ for $i = 1$ or $i = 2$ and $\mathfrak{A}_{ii} + \mathfrak{A}_{12}$ would be an ideal of $\mathfrak{A}$. From (2.5) we obtain

$$(\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{ii}) \subset (\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{12}\mathfrak{A}_{12}) \subset \mathfrak{A}_{12}\mathfrak{A}_{12} \subset \mathfrak{A}''$$

and conseqently $\mathfrak{A}' \subset \mathfrak{A}''$.

Next we discuss the subspace $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J)$ of $\mathfrak{A}$. According to the remark in § 1.**3** we know *that $g \in \mathfrak{c}$ is equivalent to*

$$g \in \mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22}), \quad g\mathfrak{A}_{12} = 0. \tag{6.3}$$

Since $e_1 - e_2$ belongs to $\mathfrak{c}$ we obtain as a consequence that the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J_c)$ has a non-trivial center $\mathfrak{C} = 0 \oplus \mathfrak{c}$.

Analogous to Lemma 8b we prove

LEMMA 12. *If $\mathfrak{A}$ and $\mathfrak{A}_{ii}$ are simple and if $\mathfrak{c} = \mathfrak{c}(\mathfrak{A}; J_c)$ then $\mathfrak{j}^+(\mathfrak{c}) = 0$ and $\mathfrak{J}^+(\mathfrak{c}) = \mathfrak{C}$.*

Proof. In order to prove $\mathfrak{j}^+(\mathfrak{c}) = 0$ we start with an element $T = L(u) + D \in \mathfrak{h}^+$ such that $T\mathfrak{A} \subset \mathfrak{c}$ and where $D \in \mathfrak{h}^+$ is an inner derivation of $\mathfrak{A}$. It follows $u = 0$ because $u \in \mathfrak{A}_{12} \cap \mathfrak{c} = 0$. From $Dc \in \mathfrak{A}_{12}$ and $D\mathfrak{A}_+ \subset \mathfrak{A}_+$ we conclude $Dc = 0$ and consequently $D\mathfrak{A}_{ij} \subset \mathfrak{A}_{ij}$. Hence $D\mathfrak{A}_{12} \subset \mathfrak{c} \subset \mathfrak{A}_+$ implies $D\mathfrak{A}_{12} = 0$ and $D\mathfrak{A}_{ii} = 0$ follows from $\mathfrak{A}_{ii} = e_i(\mathfrak{A}_{12}\mathfrak{A}_{12})$.

This completes the proof of $\mathfrak{j}^+(\mathfrak{c}) = 0$. As a consequence we have $\mathfrak{J}^+(\mathfrak{c}) = \mathfrak{j}^+(\mathfrak{c}) \oplus \mathfrak{c} = 0 \oplus \mathfrak{c}$.

3. The next Lemma will be basic for what follows.

LEMMA 13. *Let $\mathfrak{A}$ and $\mathfrak{A}_{ii}$ be simple algebras. Then for any associator ideal $\mathfrak{m}$ of the pair $(\mathfrak{A}; J_c)$ we have either $\mathfrak{A}'' \subset \mathfrak{m}$ or $\mathfrak{m} \subset \mathfrak{c}$.*

Proof. Note that for the first case we have to prove only that $\mathfrak{A}_{12} \subset \mathfrak{m}$, because then we obtain $\mathfrak{A}_{12}\mathfrak{A}_{12} = (\mathfrak{A}_{12} \nabla e)\mathfrak{A}_{12} \subset \mathfrak{h}^+\mathfrak{m} \subset \mathfrak{m}$.

Put $\mathfrak{m} = \mathfrak{m}_+ + \mathfrak{m}_{12}$, $\mathfrak{m}_+ \subset \mathfrak{A}_+ = \mathfrak{A}_{11} + \mathfrak{A}_{22}$, $\mathfrak{m}_{12} \subset \mathfrak{A}_{12}$. From (2.3) it follows that $\mathfrak{A}_{ii} \cap \mathfrak{m}_+$ is contained in the ideal $\mathfrak{a}_i = \mathfrak{a}_i(\mathfrak{m}_{12})$ of $\mathfrak{A}_{ii}$. In the case

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$\mathfrak{a}_1 = \mathfrak{A}_{11}$ or $\mathfrak{a}_2 = \mathfrak{A}_{22}$ we obtain $\mathfrak{A}_{12} = e_i\mathfrak{A}_{12} \subset \mathfrak{m}_{12}$ for $i = 1$ *or* $i = 2$ and the proof is complete. In the remaining case we consequently have

$$\mathfrak{A}_{ii} \cap \mathfrak{m}_+ = \mathfrak{a}_i(\mathfrak{m}_{12}) = 0 \text{ for } i = 1 \text{ and } i = 2. \tag{6.4}$$

According to the possibilities for the ideal $\mathfrak{b}_i = \mathfrak{b}_i(\mathfrak{m}_{12}) = e_i(\mathfrak{A}_{12}\mathfrak{m}_{12})$ of $\mathfrak{A}_{ii}$ we have to consider the following cases:

a) $\mathfrak{b}_1 = \mathfrak{b}_2 = 0$. We get $\mathfrak{A}_{12}\mathfrak{m}_{12} = 0$ and one verifies that $\mathfrak{m}_{12}$ is an ideal in $\mathfrak{A}$. Hence $\mathfrak{m}_{12} = 0$ and $\mathfrak{m} \subset \mathfrak{A}_+$. From (2.3) it follows $\mathfrak{A}_{12}\mathfrak{m} = 0$ and $(\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{m}) \subset \mathfrak{A}_{ii} \cap \mathfrak{m}$ as well as $(\mathfrak{A}_{ii}, \mathfrak{m}, \mathfrak{A}_{ii}) \subset \mathfrak{A}_{ii} \cap \mathfrak{m}$. Hence because of (6.4) the space $\mathfrak{m}$ is contained in $\mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22})$ and using $\mathfrak{A}_{12}\mathfrak{m} = 0$ and (6.3) we get $\mathfrak{m} \subset \mathfrak{c}$.

b) Exactly one $\mathfrak{b}_i$ is different from zero. Then we get $\mathfrak{A}_{ii} = \mathfrak{A}_{12}\mathfrak{m}_{12} \subset \mathfrak{m}_+$ for one i and $e_i \in \mathfrak{m}_+$ is a contradiction to (6.4).

c) $\mathfrak{b}_1 = \mathfrak{A}_{11}$ and $\mathfrak{b}_2 = \mathfrak{A}_{22}$. To any $a_1 \in \mathfrak{A}_{11}$ resp. $b_2 \in \mathfrak{A}_{22}$ using $\mathfrak{A}_{12}\mathfrak{m}_{12} \subset \mathfrak{m}_+$ we find $a_2 \in \mathfrak{A}_{22}$ resp. $b_1 \in \mathfrak{A}_{11}$ such that $a_1 + a_2 \in \mathfrak{m}_+$ resp. $b_1 + b_2 \in \mathfrak{m}_+$. From (6.4) it follows that a_2 resp. b_1 are uniquely determined by a_1 resp. b_2. Hence we obtain two injective linear maps $f: \mathfrak{A}_{11} \to \mathfrak{A}_{22}$ and $g: \mathfrak{A}_{22} \to \mathfrak{A}_{11}$ such that

$$a + f(a) \in \mathfrak{m}_+ \text{ resp. } g(b) + b \in \mathfrak{m}_+$$

for $a \in \mathfrak{A}_{11}$ resp. $b \in \mathfrak{A}_{22}$. We conclude first that $g \circ f$ resp. $f \circ g$ is the identity on $\mathfrak{A}_{11}$ resp. $\mathfrak{A}_{22}$ and secondly because of $e_i\mathfrak{m}_+ = \mathfrak{A}_{ii}$ and (6.4) that

$$\mathfrak{m}_+ = \{a + f(a)\,;\, a \in \mathfrak{A}_{11}\} \tag{6.5}$$

is satisfied. This and $(\mathfrak{A}_+, \mathfrak{m}_+, \mathfrak{A}_+) \subset \mathfrak{m}_+$ imply $(\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{ii}) \subset \mathfrak{A}_{ii} \cap \mathfrak{m}_+ = 0$. Hence $\mathfrak{A}_{11}$ and $\mathfrak{A}_{22}$ are simple associative Jordan algebras, i. e. fields.

The final step [2] for the proof of $\mathfrak{A}'' \subset \mathfrak{m}$ will be the proof of $f(e_1) = e_2$, because then we get as a consequence $e \in \mathfrak{m}_+$ and $\mathfrak{A}_{12} \subset \mathfrak{m}$. We use the quadratic representation $P(u)$ and its linearization

$$P(u, v) = L(u)L(v) + L(v)L(u) - L(uv).$$

Since $P(u, v)\mathfrak{A}_{11} \subset \mathfrak{A}_{22}$ for $u, v \in \mathfrak{A}_{12}$ holds (see for instance [1], Chapter VIII, Lemma 2.4) the formula $a(uv) + P(u, v)a = u(av) + v(au)$ yields $a(uv) + P(u, v)a \in \mathfrak{A}_{12}\mathfrak{m}_{12} \subset \mathfrak{m}_+$ for $a \in \mathfrak{A}_{11}$, $u \in \mathfrak{m}_{12}$, $v \in \mathfrak{A}_{12}$. Using (6.5) we find

$$P(u, v)a = f(ak),\ u \in \mathfrak{m}_{12},\ v \in \mathfrak{A}_{12},\ k = e_1(uv) \in \mathfrak{A}_{11}. \tag{6.6}$$

[2] There is an essential simplification for finite dimensional central simple algebras by using a field extension argument.

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In particular $P(u)a = f(ah)$, $h = e_1u^2$, for $u \in \mathfrak{m}_{12}$. From the fundamental formula $P(u)P(a)P(u) = P(P(u)a)$ it follows $[P(u)a]^2 = P(u)P(a)u^2 = P(u)[a^2h]$ because $\mathfrak{A}_{11}$ is associative. Hence $f^2(ah) = f(a^2h^2)$.

Suppose $h = e_1u^2 \neq 0$ for some $u \in \mathfrak{m}_{12}$. Choosing $a \in \mathfrak{A}_{11}$ such that $ah = e_1$ we get $f^2(e_1) = f(e_1)$ and consequently $f(e_1) = e_2$.

The proof of the Lemma will be completed if we deduce a contradiction from $h = 0$ for all $u \in \mathfrak{m}_{12}$, i.e. from $\mathfrak{m}_{12}\mathfrak{m}_{12} = 0$. Note that $\mathfrak{m}_{12}\mathfrak{A}_{12} \neq 0$. From (6.5) and $\mathfrak{m}_{12}\mathfrak{m}_{12} = 0$ it follows $([a + f(a)]u)^2 = 0$ for $a \in \mathfrak{A}_{11}$ and $u \in \mathfrak{A}_{12}$. Using the identity

$$4(uv)^2 = P(u)v^2 + P(v)u^2 + 2v[P(u)v] \tag{6.7}$$

(see for instance [1], page 146, formula 3.1) for $v = a + f(a)$ we obtain $P(u)a^2 + f^2(a)u^2 + 2f(a)[P(u)a] = 0$ by looking at the component in $\mathfrak{A}_{22}$. Linearizing and using (6.6) we get $f(a^2h_1) + f^2(a)h_2 + 2f(a)f(ah_1) = 0$ where $h_i = e_i(uv)$ and $u \in \mathfrak{m}_{12}$, $v \in \mathfrak{A}_{12}$. By linearization follows

$$f(abh_1) + f(a)f(b)h_2 + f(a)f(bh_1) + f(b)f(ah_1) = 0, \tag{6.8}$$

where $a, b \in \mathfrak{A}_{11}$. Choose $u \in \mathfrak{m}_{12}$, $v \in \mathfrak{A}_{12}$, such that $uv = h_1 + h_2 \neq 0$. Then $h_2 = f(h_1)$ and the choice $b = e_1$ resp. $b = h_1^{-1}$ yields

$$[e_2 + f(e_1)][f(ah_1) + f(a)h_2] = 0$$

resp.

$$[e_2 + f(e_1)]f(a) + f(h_1^{-1})[f(ah_1) + f(a)f(h_1)] = 0.$$

It follows that $f(ah_1) = -f(a)h_2$ and we conclude from (6.8) that

$$f(a)f(b) = -f(ab), \quad a, b \in \mathfrak{A}_{11}, \tag{6.9}$$

is fulfilled. In particular we have $f(e_1) = -e_2$.

Substituting $a = e_1$, $b = u$, $c = v$, $d = a_1 + a_2$ in (A′) and looking at the components in $\mathfrak{A}_{11}$ resp. $\mathfrak{A}_{22}$ we obtain

$$e_i(u[va_j]) = e_i(v[ua_j]) \text{ for } i \neq j \text{ and } a_i \in \mathfrak{A}_{ii}, \tag{6.10}$$

$u, v \in \mathfrak{A}_{12}$ because of (6.2).

Since $\mathfrak{m}_{12}\mathfrak{A}_{12} \neq 0$ there are $u \in \mathfrak{m}_{12}$, $v \in \mathfrak{A}_{12}$, such that $uv = k + f(k)$, $k = e_1(uv) \neq 0$. Substituting $a = k^{-1}$ in (6.6) we conclude $-e_2 = f(e_1) = f(ak) = P(u, v)a = e_2(u[va] + v[ua] - [uv]a) = 2e_2(u[va])$ because of (6.10). Replacing $2va$ by v we end up with $-e_2 = e_2(uv)$, i.e. with $uv = e_1 - e_2$ for some $u \in \mathfrak{m}_{12}$, $v \in \mathfrak{A}_{12}$. But $u^2 = (uv)v = 0$ leads to a contradiction after interchanging u and v in (6.7).

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4. The idempotent $c \neq e$ is called *decomposing for* $\mathfrak{A}$ if J_c is decomposing for $\mathfrak{A}$ in the sense of § 3. **1**.

THEOREM 6. *Let $\mathfrak{A}$ be a simple Jordan algebra and let $c \neq e$ be a decomposing idempotent for $\mathfrak{A}$ such that $\mathfrak{A}_{11}$ and $\mathfrak{A}_{22}$ are simple. Then for any ideal $\mathfrak{J}$ of he Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J_c)$ we have either $\mathfrak{J} \subset \mathfrak{C}$ or $\mathfrak{C} + \mathfrak{J} = \mathfrak{G}$.*

Proof. Forming the ideal $\mathfrak{J} = \mathfrak{k} \oplus \mathfrak{m}$ associated with $\mathfrak{J}$, where $\mathfrak{k} = H\mathfrak{J}$, $\mathfrak{m} = A\mathfrak{J}$, (see § 2. **2**) we know from the Corollary of Lemma 6 that $\mathfrak{m}$ is an associator ideal of $(\mathfrak{A}; J_c)$. Hence Lemma 13 leads to $\mathfrak{m} \subset \mathfrak{c}$ or $\mathfrak{A}'' \subset \mathfrak{m}$.

In the first case again the Corollary of Lemma 6 yields $\mathfrak{J} \subset \mathfrak{J}^+(\mathfrak{m}) \subset \mathfrak{J}^+(\mathfrak{c})$ and Lemma 12 completes the proof of $\mathfrak{J} \subset \mathfrak{C}$.

In the second case an application of (3.1) gives $\mathfrak{c} + \mathfrak{m} = \mathfrak{A}$ and consequently $A(\mathfrak{C} + \mathfrak{J}) = \mathfrak{A}$. Using Lemma 7 (for $\mathfrak{C} + \mathfrak{J}$ instead of $\mathfrak{J}$) we obtain $\mathfrak{A}'' \subset \mathfrak{C} + \mathfrak{J}$ and therefore $\mathfrak{A} = \mathfrak{c} + \mathfrak{A}'' \subset \mathfrak{C} + \mathfrak{J}$. Hence $\mathfrak{h}^+ = \mathfrak{A} \nabla \mathfrak{A} \subset \mathfrak{C} + \mathfrak{J}$ and we end p with $\mathfrak{G} = \mathfrak{C} + \mathfrak{J}$.

COROLLARY. *Suppose that moreover the sum $\mathfrak{A} = \mathfrak{c} + \mathfrak{A}''$ is direct. Then $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$ and $[\mathfrak{G}, \mathfrak{G}]$ is a simple Lie algebra.*

5. Finally we give some sufficient conditions for the idempotent to be decomposing. Denote by $\mathfrak{Z}(\mathfrak{A})$ the center of $\mathfrak{A}$. First we shall prove that for a simple Jordan algebra of finite dimension the conditions

$$\mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22}) \subset \mathfrak{c} + \mathfrak{A}_{12}\mathfrak{A}_{12}, \tag{6.11}$$

$$\mathfrak{Z}(\mathfrak{A}_{ii}) + (\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{ii}) = \mathfrak{A}_{ii} \text{ for } i = 1 \text{ or } i = 2 \text{ and} \tag{6.11'}$$

$$\mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22}) = \mathfrak{c} + \mathfrak{Z}(\mathfrak{A}) \tag{6.11''}$$

imply that c is decomposing. From the proposition in **1** we conclude $e_i(\mathfrak{A}_{12}\mathfrak{A}_{12}) = \mathfrak{A}_{ii}$ and consequently $(\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{ii}) = (\mathfrak{A}_{ii}, \mathfrak{A}_{ii}, \mathfrak{A}_{12}\mathfrak{A}_{12}) \subset \mathfrak{A}_{12}\mathfrak{A}_{12}$ because of (2.5). If (6.11′) holds for i, then (6.11) implies $\mathfrak{A}_{ii} \subset \mathfrak{c} + \mathfrak{A}_{12}\mathfrak{A}_{12}$ and from $\mathfrak{A}_{jj} = e_j(\mathfrak{A}_{12}\mathfrak{A}_{12}) \subset \mathfrak{A}_{12}\mathfrak{A}_{12} + \mathfrak{A}_{ii} \subset \mathfrak{c} + \mathfrak{A}_{12}\mathfrak{A}_{12}$ it follows the first condition in (3.1′). But its second condition is the same as (6.11″) because $\mathfrak{Z}(\mathfrak{A}) \subset \mathfrak{A}_+$.

Let r be the degree of the central simple algebra $\mathfrak{A}$ and let p be the characteristic of the ground field K. Then $\mathfrak{A} = \mathfrak{Z}(\mathfrak{A}) + (\mathfrak{A}, \mathfrak{A}, \mathfrak{A})$ whenever r is not divisible by p. This result is due to K. H. Helwig [2]. In the case of $p = 0$ (or p big enough) the result is proved by R. D. Schafer [8] and K. Meyberg [7].

THEOREM 7. *Let $\mathfrak{A}$ be a finite dimensional central simple Jordan algebra*

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over the field K of characteristic p and let $c \neq e$ be an idempotent of $\mathfrak{A}$. Suppose that p is not a divisor of the degree r of $\mathfrak{A}$.

Then c is directly decomposing and the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J_c)$ is a direct sum of its center $\mathfrak{C}$ and the simple algebra $[\mathfrak{G}, \mathfrak{G}]$.

Proof. We have only to prove that J_c is decomposing because of § 3. 2, Theorem 2 and the Corollary to Theorem 6. Since being decomposing is invariant under ground field extension we may assume that K is algebraically closed. There exists an integer s, $0 < s < r$, such that $\mathfrak{A}_{11}$ resp. $\mathfrak{A}_{22}$ has the degree s resp. $r - s$. Not both can be divisible by p, therefore without restriction we may assume that p is not a divisor of s. By Helwig's result we have $\mathfrak{A}_{11} = Ke_1 + (\mathfrak{A}_{11}, \mathfrak{A}_{11}, \mathfrak{A}_{11})$ i. e. (6. 11′) is true.

Since $\mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22})$ is spanned by e_1 and e_2 it follows from (6. 3) that c is spanned by $e_1 - e_2$. Hence (6. 11″) is fulfilled and we have $\mathfrak{Z}(\mathfrak{A}_{11}) + \mathfrak{Z}(\mathfrak{A}_{22}) = Ke + K(e_1 - e_2)$. According to (6. 11) it suffices to prove

$$e \in K(e_1 - e_2) + \mathfrak{A}_{12}\mathfrak{A}_{12}. \tag{6. 12}$$

In the case $r = 2$, i. e. if $\mathfrak{A}$ is given by a non degenerate bilinear form in a vector space, we have $\mathfrak{A}_{12}\mathfrak{A}_{12} = Ke$.

In order to prove (6. 12) we may assume $r \geqq 3$. Let $\mathfrak{H}_r$ be the Jordan algebra over K of symmetric $r \times r$ matrices $x = (x_{ij})$, $x_{ij} \in K$. For a given integer s, $0 < s < r$, we write the elements x of $\mathfrak{H}_r$ in the form

$$x = \begin{pmatrix} u & v^t \\ v & w \end{pmatrix}, \quad u \in \mathfrak{H}_s, \quad w \in \mathfrak{H}_{r-s},$$

where v is an arbitrary $(r - s) \times s$ matrix. We use the abbreviation $x = [u, w]$ whenever $v = 0$. Denote the unit matrix in $\mathfrak{H}_r$ by 1_r.

Using the classification of simple Jordan algebra over an algebraically closed ground field and changing if necesssary from $\mathfrak{A}$ to an isomorphic image we may assume that $\mathfrak{H}_r$ is a subalgebra of $\mathfrak{A}$ and that $c = [1_s, 0]$ is the given idempotent. Therefore it is enough to prove (6. 12) for $\mathfrak{A} = \mathfrak{H}_r$. Furthermore we may assume $2s < r$, because $2s = r$ implies $e \in \mathfrak{A}_{12}\mathfrak{A}_{12}$, and we may change s into $r - s$.

Let $\mathfrak{D}$ be the set of diagonal matrices in $K(e_1 - e_2) + \mathfrak{A}_{12}\mathfrak{A}_{12}$. The elements of $\mathfrak{A}_{12}\mathfrak{A}_{12}$ turn out to be linear combinations of matrices $[v^t v, vv^t]$ where v are $(r - s) \times s$ matrices. Taking $v^t = (1_s, 0)$ we obtain $[1_{2s}, 0] \in \mathfrak{D}$, moreover $[1_s, -1_{r-s}] \in \mathfrak{D}$. On the other side taking for v only matrices which have non-zero entries only in one colmn, we get $[u(w), w] \in \mathfrak{D}$ whenever

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w is diagonal and $u(w)$ is the diagonal matrix having the trace of w at a given position in the diagonal and zero elsewhere. But one verifies that e is a linear combination of $[1_{2s}, 0]$, $[1_s, -1_{r-s}]$ and $[u(w), w]$, w diagonal, provided p is not a divisor of r.

7. Applications to formal real Jordan algebras.

1. Let $\mathfrak{A}$ be a *formal real* Jordan algebra, i.e. a finite dimensional Jordan algebra over the field of real numbers such that $u^2 + v^2 = 0$ implies $u = v = 0$. For the results about formal real Jordan algebras used in the sequel see Chapter XI in [1]. The bilinear form λ given by

$$\lambda(a, b) = \operatorname{Trace} L(ab), \quad a, b \in \mathfrak{A},$$

turns out to be a positive definite bilinear form of $\mathfrak{A}$. Furthermore $\mathfrak{A}$ is semi simple and is a direct sum of central simple ideals. Let J be an automorphism of $\mathfrak{A}$ satisfying $J^2 = \mathrm{Id}$.

LEMMA 14. *The bilinear form* $\delta(a, b) = \operatorname{Trace}\{L(ab) - P(a, Jb)J\}$ *of* $\mathfrak{A}$ *is positive semi definite.*

Proof. By the same arguments as used in the proof of Theorem 5 we may assume tha $(\mathfrak{A}; J)$ is a simple pair. Then we have to discuss the three cases:

a) The pair is of type I: Then $\delta(a, b) = \lambda(a, b)$.

b) The pair is of Type II and $J \neq \mathrm{Id}$: Then f is a non-zero element in $\mathfrak{Z}(\mathfrak{A})$, i.e. $f = \alpha e$ and $\delta(e, e) = \operatorname{Trace}\ (\mathrm{Id} - J) > 0$ yields $\alpha > 0$.

c) The pair is of type II and $J = \mathrm{Id}$: Then $\delta = 0$.

From Theorem 5a) we get the

COROLLARY. *If* J *is a decomposing automorphism for* $\mathfrak{A}$ *then the restriction of* δ *to* $\mathfrak{A}'' = \mathfrak{A}_-\mathfrak{A}_- + \mathfrak{A}_-$ *is positive definite and* δ *is zero on* $\mathfrak{c}$.

2. Let $\mathfrak{d} = \mathfrak{d}(\mathfrak{A}) = [L(\mathfrak{A}), L(\mathfrak{A})]$ be the Lie algebra of (inner) derivations of $\mathfrak{A}$. Then $D^* = -D$ where D^* stands for the adjoint of D with respect to λ. The remark in § 5. **1** gives

$$\mathfrak{d}^+ = [L(\mathfrak{A}_+), L(\mathfrak{A}_+)] + [L(\mathfrak{A}_-), L(\mathfrak{A}_-)].$$

Calling the zero-form on the zero-space "negative definite" we have the

LEMMA 15. *Let* J *be a decomposing automorphism of* $\mathfrak{A}$. *Then the Killing form of* $\mathfrak{d}^+$ *is negative definite, i.e.* $\mathfrak{d}^+$ *is a compact semi simple Lie algebra or* 0.

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Proof. We may restrict ourselves to the case that $\mathfrak{A}$ is a simple algebra. Suppose first that $\mathfrak{A}$ is not isomorphic to the algebra of 2×2 symmetric matrices. Then the Killing form $\langle D_1, D_2 \rangle_{\mathfrak{d}}$ of $\mathfrak{d}$ coincides up to a positive factor with Trace $D_1 D_2$ (see [1], Chapter IX, Satz 3.6). Hence $\langle D, D \rangle_{\mathfrak{d}}$ is negative for $0 \neq D \in \mathfrak{d}$, and the restriction of $\langle\ ,\ \rangle_{\mathfrak{d}}$ to $\mathfrak{d}^+$ turns out to be a negative definite associative bilinear form of $\mathfrak{d}^+$. By standard arguments we conclude that $\mathfrak{d}^+$ is a compact semi simple Lie algebra having a negative definite Killing form.

Now let $\mathfrak{A}$ be the algebra of 2×2 symmetric matrices. Then $\mathfrak{d}$ has dimension one (see [1], Chapter IX, Satz 3.3). Hence $\mathfrak{d}^+ = 0$ or $\mathfrak{d}^+ = \mathfrak{d}$ and $\mathfrak{d}^- = 0$. But $L(\mathfrak{A}_-) = \mathfrak{d}^- = 0$ would imply $J = \mathrm{Id}$ and J were not decomposing.

3. We use the decompositions

$$\mathfrak{G} = \mathfrak{K}_\epsilon + \mathfrak{P}_\epsilon, \quad \mathfrak{K}_\epsilon = \mathfrak{d}^+ \oplus \mathfrak{A}_\epsilon, \quad \mathfrak{P}_\epsilon = L(\mathfrak{A}_-) + \mathfrak{A}_{-\epsilon}, \quad \epsilon = \pm 1,$$

(see § 5. **1**) of the Lie algebra $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ according to the automorphism Θ_ϵ, $\epsilon = \pm 1$. The elements of $\mathfrak{G}$ are written as

$$x = (D + L(d)) \oplus (a_+ + a_-), \ D \in \mathfrak{d}^+, \ \ d \in \mathfrak{A}_-, \ a_\epsilon \in \mathfrak{A}_\epsilon,$$

and we use the abbreviations

$$x_\epsilon^+ = D \oplus a_\epsilon \in \mathfrak{K}_\epsilon, \quad x_\epsilon^- = L(d) \oplus a_{-\epsilon} \in \mathfrak{P}_\epsilon,$$

as well as $T = D + L(d) \in \mathfrak{h}^+$ and $a_+ = a_+ + a_-' \in \mathfrak{A}$. Using Lemma 10 and (5.1) we obtain

$$(7.1) \qquad \langle x, x \rangle_{\mathfrak{G}} = \langle D, D \rangle_{\mathfrak{d}^+} + \mathrm{Trace}\, D^2 + \mathrm{Trace}_-\,(D^2 + 2L(d^2)) + 4\delta(a, Ja)$$

and consequently

$$(7.2) \qquad \langle x_\epsilon^+, x_\epsilon^+ \rangle_{\mathfrak{G}} = \langle D, D \rangle_{\mathfrak{d}^+} + \mathrm{Trace}\, D^2 + \mathrm{Trace}_-\, D^2 + 4\epsilon\delta(a_\epsilon, a_\epsilon),$$

$$(7.3) \qquad \langle x_\epsilon^-, x_\epsilon^- \rangle_{\mathfrak{G}} = 2\,\mathrm{Trace}_-\, L(d^2) - 4\epsilon\delta(a_{-\epsilon}, a_{-\epsilon}).$$

THEOREM 8. *Let $\mathfrak{A}$ be a formal real Jordan algebra and let J, $J^2 = \mathrm{Id}$, be a decomposing automorphism of $\mathfrak{A}$. Then*

(i) *$\mathfrak{K}_-$ is a maximal compact semi simple subalgebra of $\mathfrak{G} = \mathfrak{G}(\mathfrak{A}; J)$ and of $[\mathfrak{G}, \mathfrak{G}]$.*

(ii) *$\mathfrak{G} = \mathfrak{K}_- + \mathfrak{P}_-$ is a Cartan decomposition of $\mathfrak{G}$, i.e. the Killing form of $\mathfrak{G}$ is negative definite on $\mathfrak{K}_-$ and positive semi definite on $\mathfrak{P}_-$.*

(iii) *$[\mathfrak{G}, \mathfrak{G}] = \mathfrak{K}_- + [\mathfrak{K}_-, \mathfrak{P}_-]$ is a Cartan decomposition of the semi simple algebra $[\mathfrak{G}, \mathfrak{G}]$.*

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We divide the *proof* in the following steps:

a) *The Killing form of* $\mathfrak{G}$ *is negative semi definite on* $\mathfrak{K}_-$: We use (7.2) for $\epsilon = -1$. Because of Lemma 14 and Lemma 15 it is enough to prove that $\operatorname{Trace} D^2 + \operatorname{Trace}_- D^2$ is negative or zero for $D \in \mathfrak{d}^+$. But $D^* = -D$ shows that the transformation $D^2 = -D^*D$ is negative semi definite with respect to the positive definite bilinear form λ. Hence the trace of the restriction of D^2 to the invariant subspaces $\mathfrak{A}$ resp. $\mathfrak{A}_-$ is negative or zero.

b) *The Killing form of* $\mathfrak{G}$ *is positive semi definite on* $\mathfrak{P}_-$: Because of (7.3) and Lemma 14 it is enough to prove that the linear transformation $L(u^2)$ for $u \in \mathfrak{A}$ is positive semi definite. But this is a consequence of [1], Chapter XI, Satz 3.6b together with Satz 3.7b.

c) *Proof of* (ii): The spaces $\mathfrak{K}_-$ and $\mathfrak{P}_-$ are orthogonal with respect to the Killing form of $\mathfrak{G}$. Moreover the bilinear kernel of the Killing form of $\mathfrak{G}$, i.e. the center $\mathfrak{C} = 0 \oplus \mathfrak{c}$ of $\mathfrak{G}$ (see Theorem 5) is contained in $\mathfrak{P}_-$. Hence the Killing form has to be non degenerate on $\mathfrak{K}_-$ and the statement follows from a) and b).

d) *The Killing form of* $\mathfrak{K}_-$ *is negative definite*: From (ii) it follows that the restriction of the Killing form of $\mathfrak{G}$ to $\mathfrak{K}_-$ is a negative definite associative bilinear form of $\mathfrak{K}_-$. Therefore the Killing form of $\mathfrak{K}_-$ has to be negative definite.

e) $[\mathfrak{G}, \mathfrak{G}] = \mathfrak{K}_- + [\mathfrak{K}_-, \mathfrak{P}_-]$ *and* $\mathfrak{P}_- = \mathfrak{C} + [\mathfrak{K}_-, \mathfrak{P}_-]$: Part d) implies that $\mathfrak{K}_-$ is semi simple and hence that $\mathfrak{K}_- = [\mathfrak{K}_-, \mathfrak{K}_-]$. Therefore the statement follows from $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$ (see Corollary 2 of Theorem 5).

f) *Proof of* (i) *and* (iii): Since the Killing form of $[\mathfrak{G}, \mathfrak{G}]$ is the restriction of the Killing form of $\mathfrak{G}$ to $[\mathfrak{G}, \mathfrak{G}]$ we observe (iii) from (ii) and part e). Hence $\mathfrak{K}_-$ is a maximal compact semi simple subalgebra of $[\mathfrak{G}, \mathfrak{G}]$. The analogous statement for $\mathfrak{G}$ follows from $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G}, \mathfrak{G}]$.

4. Next we consider the subalgebra $\mathfrak{K}_+$ and the decomposition $\mathfrak{G} = \mathfrak{K}_+ + \mathfrak{P}_+$.

LEMMA 16. *Let J be a decomposing automorphism for $\mathfrak{A}$. Then*

(i) $\operatorname{center} \mathfrak{K}_+ = 0 \oplus \mathfrak{Z}(\mathfrak{A}_+)$.

(ii) $[\mathfrak{K}_+, \mathfrak{K}_+] = \mathfrak{d}^+ \oplus \mathfrak{d}^+\mathfrak{A}_+ \subset \mathfrak{d}^+ \oplus \mathfrak{A}_-\mathfrak{A}_-$.

(iii) $\mathfrak{K}_+ = \operatorname{center} \mathfrak{K}_+ + [\mathfrak{K}_+, \mathfrak{K}_+]$.

(iv) $[\mathfrak{P}_+, \mathfrak{P}_+] = [L(\mathfrak{A}_-), L(\mathfrak{A}_-)] \oplus \mathfrak{A}_-\mathfrak{A}_-$.

Proof. (i) Let $C \oplus c$ be an element in the center of $\mathfrak{K}_+$. Hence

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$[C, D] + c \nabla a = 0$ and $Ca = Dc$ for $D \in \mathfrak{d}^+$ and $a \in \mathfrak{A}_+$. In particular $C \in$ center $\mathfrak{d}^+$ and Lemma 15 leads to $C = 0$. We obtain $c \nabla \mathfrak{A}_+ = \mathfrak{d}^+ c = 0$. Hence c belongs to the center of $\mathfrak{R}_+$ if and only if both $[L(c), L(\mathfrak{A}_+)] = 0$ and $(\mathfrak{A}_\epsilon, c, \mathfrak{A}_\epsilon) = 0$ for $\epsilon = \pm 1$ is satisfied. In particular we get $c \in \mathfrak{Z}(\mathfrak{A}_+)$. Conversely, let be $c \in \mathfrak{Z}(\mathfrak{A}_+)$. From (3.1) it follows $c = g + z$ where $g \in \mathfrak{c}$ and $z \in \mathfrak{Z}(\mathfrak{A})$, and we deduce $c \in$ center $\mathfrak{R}_+$ from Lemma 1.

(ii) From $\mathfrak{R}_+ = \mathfrak{d}^+ \oplus \mathfrak{A}_+$ we obtain

$$[\mathfrak{R}_+, \mathfrak{R}_+] = ([\mathfrak{d}^+, \mathfrak{d}^+] + \mathfrak{A}_+ \nabla \mathfrak{A}_+) \oplus \mathfrak{d}^+\mathfrak{A}_+ = \mathfrak{d}^+ \oplus \mathfrak{d}^+\mathfrak{A}_+$$

because $\mathfrak{A}_+ \nabla \mathfrak{A}_+ \subset \mathfrak{d}^+$ and $[\mathfrak{d}^+, \mathfrak{d}^+] = \mathfrak{d}^+$ (see Lemma 15). Furthermore

$$\begin{aligned}\mathfrak{d}^+\mathfrak{A}_+ &= \mathfrak{d}^+(\mathfrak{c} + \mathfrak{A}_-\mathfrak{A}_-) \\ &= \mathfrak{d}^+(\mathfrak{A}_-\mathfrak{A}_-) \subset (\mathfrak{A}_+, \mathfrak{A}_-\mathfrak{A}_-, \mathfrak{A}_+) + (\mathfrak{A}_-, \mathfrak{A}_-\mathfrak{A}_-, \mathfrak{A}_-) \subset \mathfrak{A}_-\mathfrak{A}_-\end{aligned}$$

because of (2.5).

(iii) Since $\mathfrak{A}_+$ is semi simple we obtain $\mathfrak{A}_+ = \mathfrak{Z}(\mathfrak{A}_+) + (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_+)$ (see R. D. Schafer [8] or K. Meyberg [7] and hence $\mathfrak{A}_+ \subset \mathfrak{Z}(\mathfrak{A}_+) + \mathfrak{d}^+\mathfrak{A}_+$. Therefore from (i) and (ii) it follows

$$\mathfrak{R}_+ = \mathfrak{d}^+ \oplus \mathfrak{A}_+ \subset \mathfrak{Z}(\mathfrak{A}_+) + (\mathfrak{d}^+ \oplus \mathfrak{d}^+\mathfrak{A}_+) \subset \text{center } \mathfrak{R}_+ + [\mathfrak{R}_+, \mathfrak{R}_+].$$

(iv) We have

$$\begin{aligned}[\mathfrak{P}_+, \mathfrak{P}_+] &= [L(\mathfrak{A}_-) \oplus \mathfrak{A}_-, L(\mathfrak{A}_-) \oplus \mathfrak{A}_-] \\ &= ([L(\mathfrak{A}_-), L(\mathfrak{A}_-)] + \mathfrak{A}_- \nabla \mathfrak{A}_-) \oplus \mathfrak{A}_-\mathfrak{A}_-\end{aligned}$$

and the statement follows from (2.1).

5. Let J be decomposing for $\mathfrak{A}$. The subspace

$$\mathfrak{I} = \mathfrak{d}^+ \oplus \mathfrak{A}_-\mathfrak{A}_- \tag{7.4}$$

of $\mathfrak{R}_+$ turns out o be a subalgebra. Indeed, it follows $[\mathfrak{I}, \mathfrak{I}] \subset \mathfrak{d}^+ \oplus \mathfrak{d}^+(\mathfrak{A}_-\mathfrak{A}_-) \subset \mathfrak{I}$ because of $\mathfrak{d}^+(\mathfrak{A}_-\mathfrak{A}_-) \subset \mathfrak{d}^+\mathfrak{A}_+ \subset \mathfrak{A}_-\mathfrak{A}_-$ (see Lemma 16 ii). Since J is decomposing we obtain

$$\mathfrak{R}_+ = \mathfrak{C} + \mathfrak{I}, \tag{7.5}$$

where $\mathfrak{C} = 0 \oplus \mathfrak{c}$ is the center of $\mathfrak{G}$. In order to prove

$$\text{center } \mathfrak{I} = 0 \oplus \mathfrak{Z}(\mathfrak{A}_+) \cap \mathfrak{A}_-\mathfrak{A}_- \tag{7.6}$$

we observe from Lemma 16 i that the right hand side is contained in center $\mathfrak{I}$. And the converse is a consequence of (7.5). Next we prove

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$$\mathfrak{k} = \text{center}\,\mathfrak{k} + [\mathfrak{K}_+, \mathfrak{K}_+]. \tag{7.7}$$

In view of (7.6) and Lemma 16 ii it suffices to prove $\mathfrak{A}_+ = \mathfrak{Z}(\mathfrak{A}_+) + \mathfrak{d}^+\mathfrak{A}_+$. But this is implied by $(\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_+) \subset \mathfrak{d}^+\mathfrak{A}_+$ and $\mathfrak{A}_+ = \mathfrak{Z}(\mathfrak{A}_+) + (\mathfrak{A}_+, \mathfrak{A}_+, \mathfrak{A}_+)$. The second condition in (3.1) yields $\mathfrak{Z}(\mathfrak{A}_+) = \mathfrak{c} + \mathfrak{Z}(\mathfrak{A}) \cap \mathfrak{A}_+$. Hence

$$\mathfrak{Z}(\mathfrak{A}_+) \cap \mathfrak{d}^+\mathfrak{A}_+ = 0 \tag{7.8}$$

follows from $\mathfrak{c} \cap \mathfrak{d}^+\mathfrak{A}_+ \subset \mathfrak{c} \cap \mathfrak{A}_-\mathfrak{A}_- = 0$ together with

$$\mathfrak{Z}(\mathfrak{A}) \cap \mathfrak{d}^+\mathfrak{A}_+ \subset \mathfrak{Z}(\mathfrak{A}) \cap (\mathfrak{A}, \mathfrak{A}, \mathfrak{A}) = 0.$$

As a consequence of (7.8) we obtain *that* (7.7) *is a direct sum decomposition.*

We recall the decomposition

$$\mathfrak{P}_+ = \mathfrak{p}^+ + \mathfrak{p}^-$$

where $\mathfrak{p}^\eta = \{2L(w) \oplus \eta;\, w \in \mathfrak{A}_-\}$ are abelian subalgebras of $\mathfrak{G}$ (see the remark in § 5.1). In order to prove

$$[\mathfrak{K}_+, \mathfrak{p}^\eta] \subset \mathfrak{p}^\eta \quad (\eta = \pm 1) \tag{7.9}$$

one verifies using (2.1) that for $D \in \mathfrak{d}^+$, $a \in \mathfrak{A}_+$, $w \in \mathfrak{A}_-$,

$$\begin{aligned}[D \oplus a, 2L(w) \oplus \eta w] &= (2[D, L(w)] + \eta a \,\nabla\, w) \oplus \eta(Dw - 2\eta wa) \\ &= 2L(v) \oplus \eta v, \text{ where } v = Dw - 2\eta wa \in \mathfrak{A}_-,\end{aligned}$$

is satisfied.

Furthermore we recall the automorphism Θ_- of $\mathfrak{G}$ (see § 1.4) defined by

$$\Theta_- x = -(T^* \oplus Ja), \text{ whenever } x = T \oplus a \in \mathfrak{G} = \mathfrak{h}^+ \oplus \mathfrak{A}.$$

Denote by Θ the restriction of Θ_- to $[\mathfrak{G}, \mathfrak{G}]$.

THEOREM 9. *Let $\mathfrak{A}$ be a formal real Jordan algebra and let J, $J^2 = \mathrm{Id}$, be a decomposing automorphism of $\mathfrak{A}$. Then*

$$[\mathfrak{G}, \mathfrak{G}] = \mathfrak{k} + \mathfrak{p}^+ + \mathfrak{p}^-$$

is a direct sum decomposition of the semi simple Lie algebra $[\mathfrak{G}, \mathfrak{G}]$ having the following properties:

(i) *$\mathfrak{p}^\eta$ $(\eta = \pm 1)$ are abelian subalgebras of $[\mathfrak{G}, \mathfrak{G}]$.*

(ii) *$[\mathfrak{k}, \mathfrak{p}^\eta] \subset \mathfrak{p}^\eta$ for $\eta = \pm 1$.*

(iii) *Θ is an automorphism of $[\mathfrak{G}, \mathfrak{G}]$ satisfying $\Theta^2 = \mathrm{Id}$, $\Theta\mathfrak{k} = \mathfrak{k}$ and $\Theta\mathfrak{p}^+ = \mathfrak{p}^-$.*

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(iv) $[\mathfrak{G},\mathfrak{G}] = \mathfrak{k} + \mathfrak{p}$ *is a Cartan decomposition of* $[\mathfrak{G},\mathfrak{G}]$ *where*
$\mathfrak{k} = \{x; x \in [\mathfrak{G},\mathfrak{G}], \Theta x = x\} = \mathfrak{K}_-$
$\mathfrak{p} = \{x; x \in [\mathfrak{G},\mathfrak{G}], \Theta x = -x\} = [\mathfrak{K}_-, \mathfrak{P}_-]$.

Proof. Part ii and iv of Lemma 16 leads to $[\mathfrak{K}_+, \mathfrak{K}_+] + [\mathfrak{P}_+, \mathfrak{P}_+] = \mathfrak{l}$. Hence using $\mathfrak{G} = \mathfrak{C} + [\mathfrak{G},\mathfrak{G}] = \mathfrak{K}_+ + \mathfrak{P}_+$ we obtain $[\mathfrak{G},\mathfrak{G}] = \mathfrak{l} + [\mathfrak{K}_+, \mathfrak{P}_+]$ and therefore $\mathfrak{G} = \mathfrak{C} + \mathfrak{l} + [\mathfrak{K}_+, \mathfrak{P}_+]$. Since $\mathfrak{C}$ and $\mathfrak{l}$ are subspaces of $\mathfrak{K}_+$ we end up with $[\mathfrak{K}_+, \mathfrak{P}_+] = \mathfrak{P}_+$ and the decomposition $[\mathfrak{G},\mathfrak{G}] = \mathfrak{l} + \mathfrak{p}^+ + \mathfrak{p}^-$. We know already that (i) holds. (ii) follows from (7.9) and $\mathfrak{l} \subset \mathfrak{K}_+$, and (iii) follows from (2.7) and (7.4). Finally we deduce (iv) from part iii of Theorem 8.

6. *Final remarks.* According to Theorem 9 we obtain a decomposition of $[\mathfrak{G},\mathfrak{G}]$ that is characteristic for the Lie algebras of bounded symmetric domains, i. e. of Hermitian symmetric spaces of noncompact type. Conversely, using a recent result of U. Hirzebruch [3] one can show that one covers all Lie algebras of bounded symmetric domains by this construction. In fact, for the irreducible domains one needs the following three classes of pairs $(\mathfrak{A}; J)$:

(A) $\mathfrak{A}$ is the direct sum $\mathfrak{B} \oplus \mathfrak{B}$ where $\mathfrak{B}$ is a simple formal real Jordan algebra and where J is the canonical automorphism.

(B) $\mathfrak{A}$ is the Jordan algebra of complex herimitian $r \times r$ matrices $(r > 2)$ and J transposes the matrices.

(C) $\mathfrak{A}$ is a simple formal real Jordan algebra and $J = J_c$ is an automorphism defined by an idempotent c of $\mathfrak{A}$.

A proof will be given elsewhere.

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